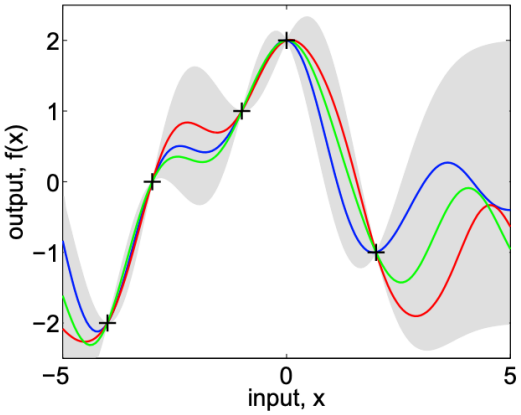
FIT5201 - Machine Learning

Module 6: Advanced Topics in Machine Learning

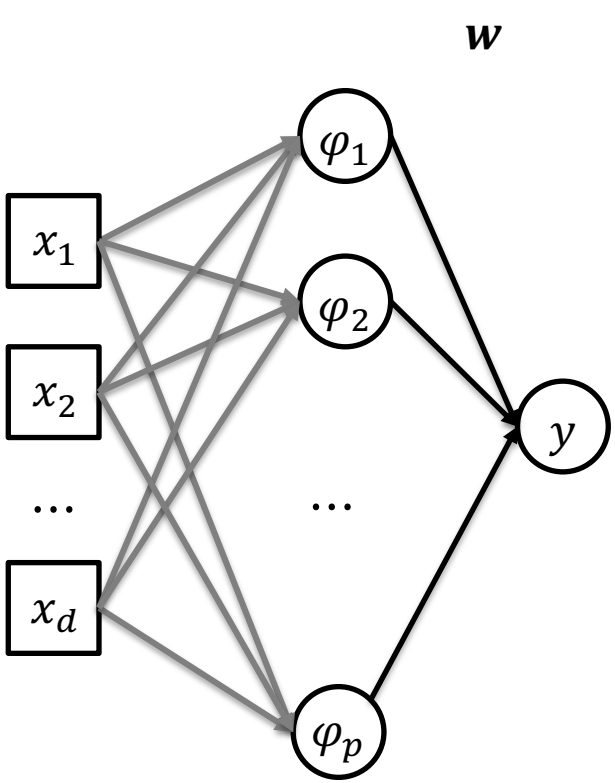
Parametric versus Non-Parametric Models

Bayesian Prediction and Model Selection

Gaussian Process Regression (GPR)



Recap: Linear Models



input representation by fixed set of basis functions (features)

$$\varphi_1, \dots \varphi_p$$

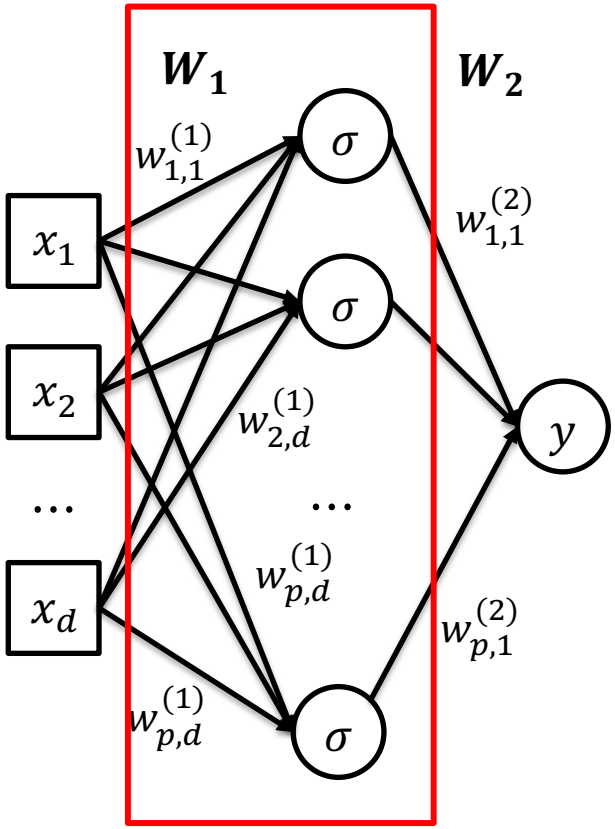
form prediction linear in learnable parameters (or transformation of linear discriminant function for classification)

$$y = \boldsymbol{w}^T \boldsymbol{\phi}(\boldsymbol{x})$$

find maximum likelihood / minimum error parameter values

$$\boldsymbol{w}^* = \underset{\boldsymbol{w} \in \mathbb{R}^p}{\operatorname{argmin}} E(\boldsymbol{w})$$

Recap: Neural Networks



$$\varphi_j(\boldsymbol{x}) = \sigma(\boldsymbol{W_1 x})$$

input representation by learnable set of basis functions (features)

$$\varphi_1, \dots \varphi_p$$

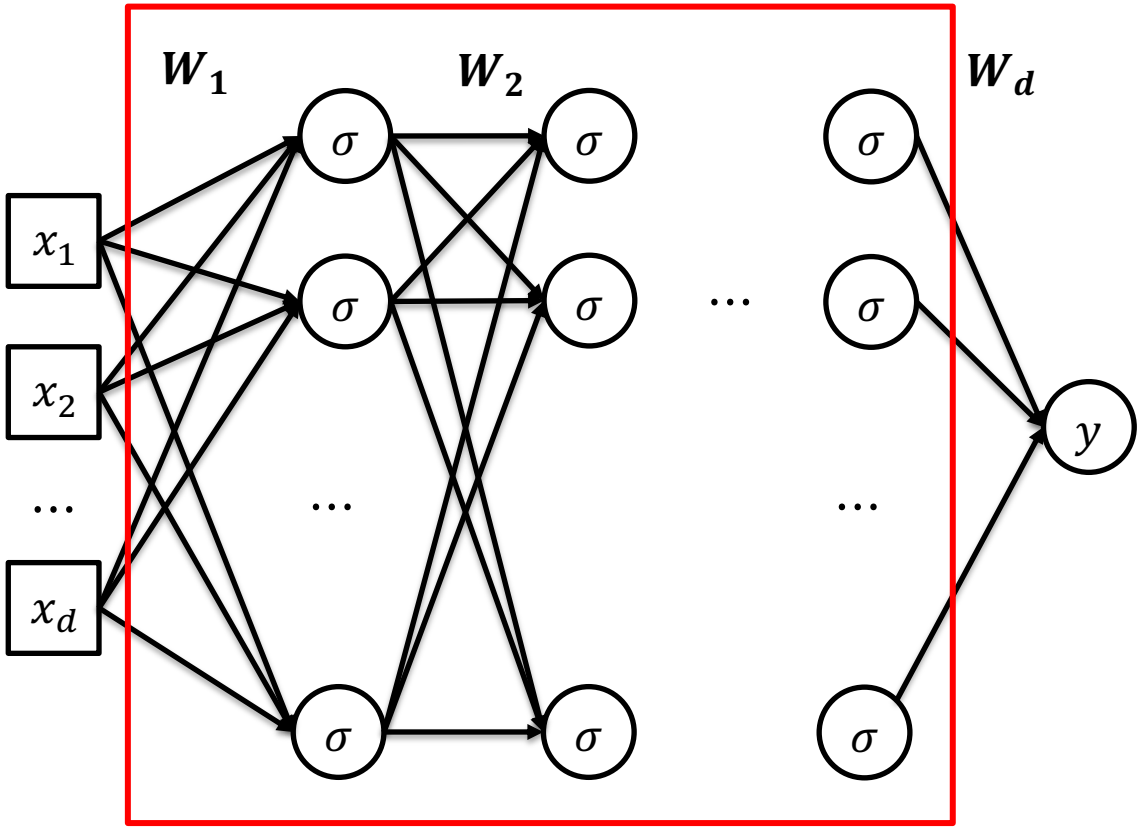
form prediction non-linear in learnable parameters (or transformation of linear discriminant function for classification)

$$y = \boldsymbol{W}_2 \sigma(\boldsymbol{W}_1 \boldsymbol{x})$$

approximate maximum likelihood / minimum error parameter values

$$\boldsymbol{W}^* = \underset{\boldsymbol{W} \in \mathbb{R}^{p \times d+p}}{\operatorname{argmin}} \; E(\boldsymbol{W})$$

Recap: Deep Neural Networks



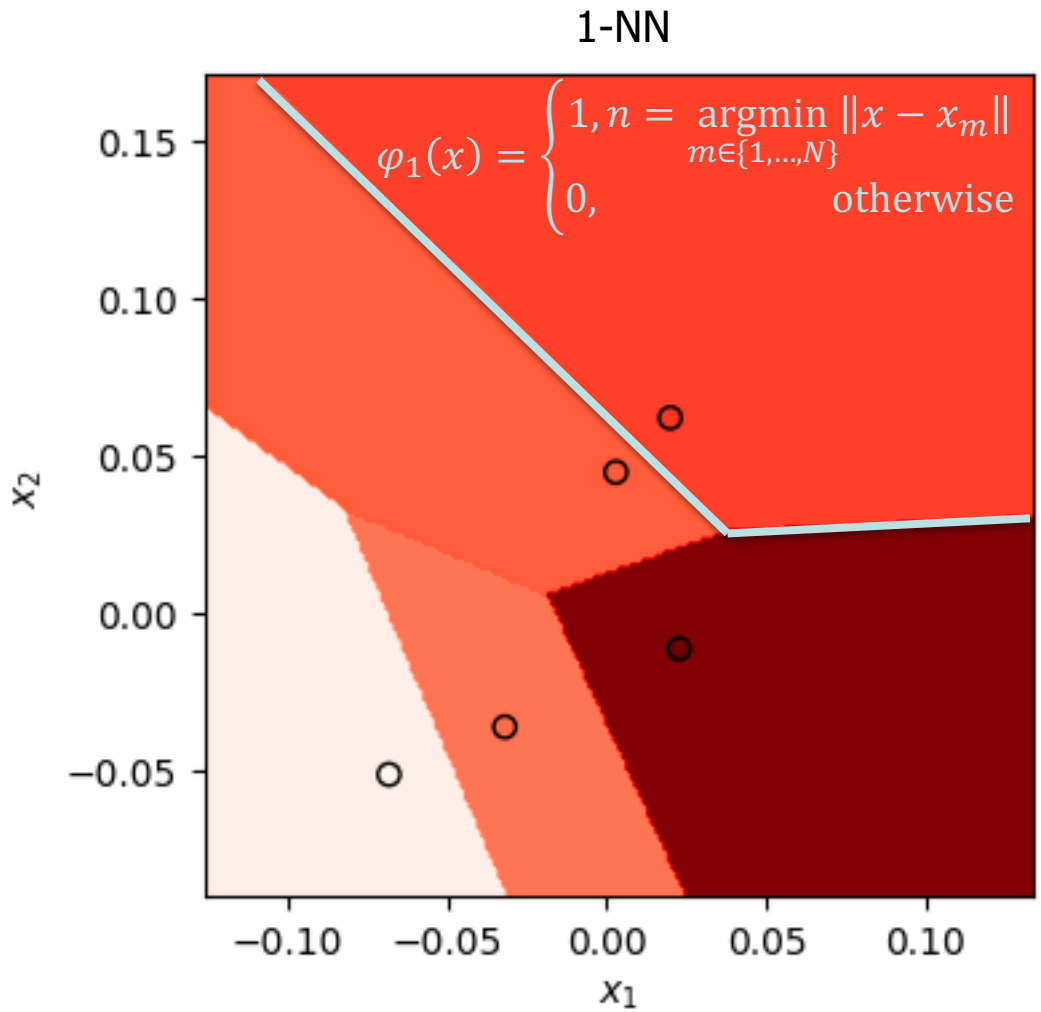
$$\varphi_j(\boldsymbol{x}) = \sigma\left(\boldsymbol{W_d}\sigma\left(\boldsymbol{W_{d-1}}\sigma\left(\ldots\sigma(\boldsymbol{W_1}\boldsymbol{x})\right)\right)\right)$$

Again fixed set of parameters to be optimized.

What do we mean by "fixed"?

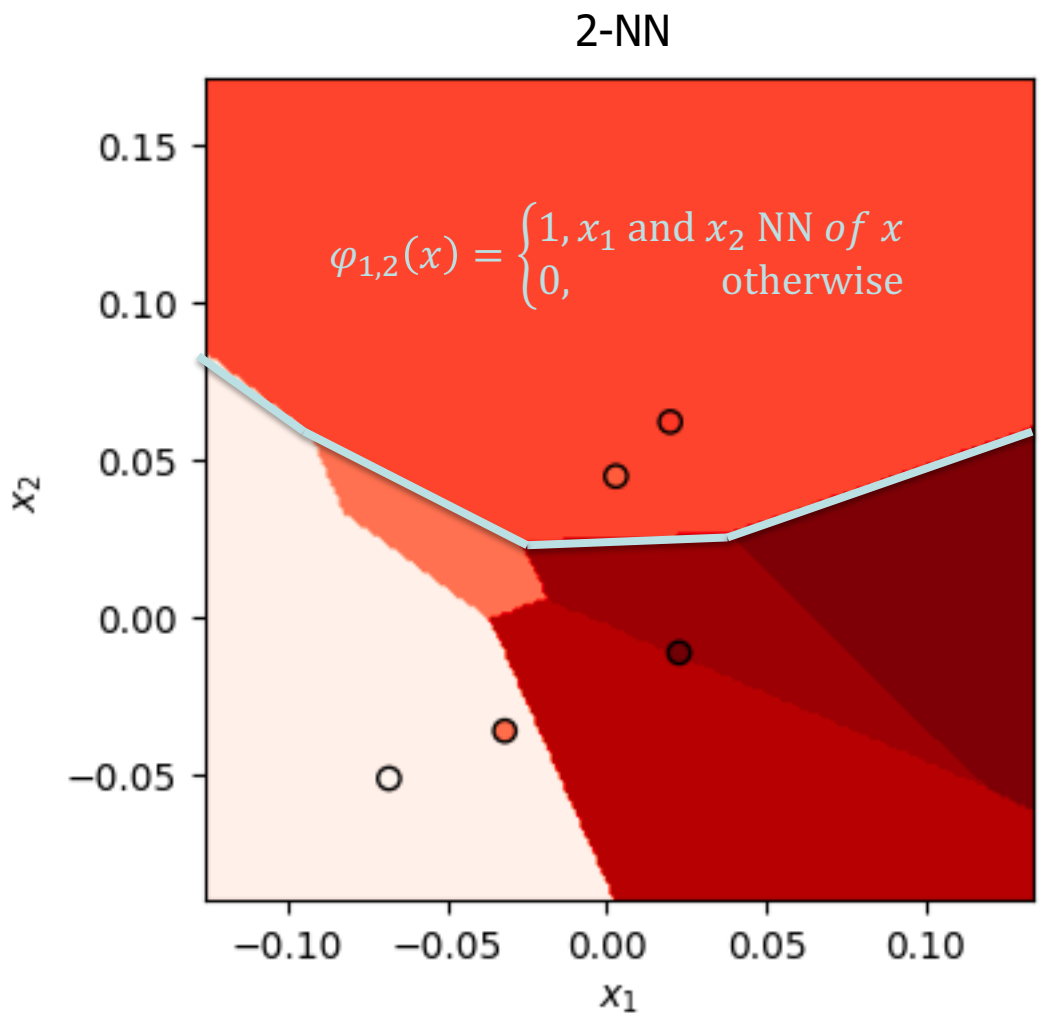
Fixed prior to observing any data, i.e. independent of data and number of data points.

Contrast: K-Nearest Neighbour



Could still write prediction as linear combination of basis functions

Contrast: K-Nearest Neighbour



Could still write prediction as linear combination of basis functions

However, number and definition of basis functions depend on data

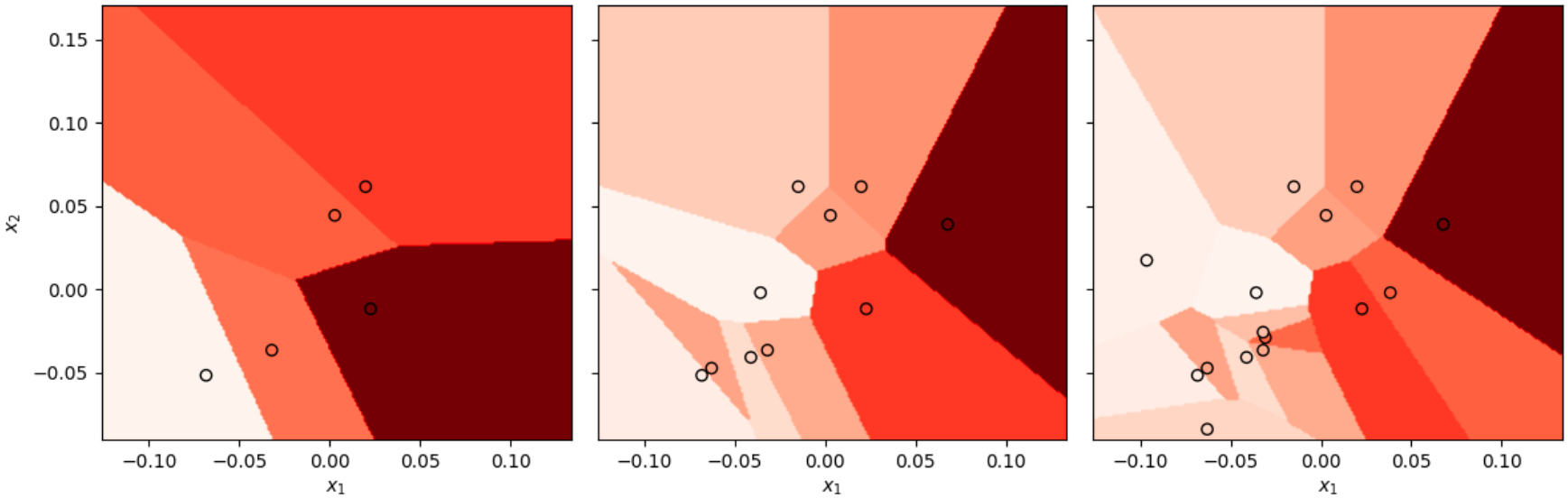
Non-Parametric Model Characteristics

Increasing number of basis functions with number of data points

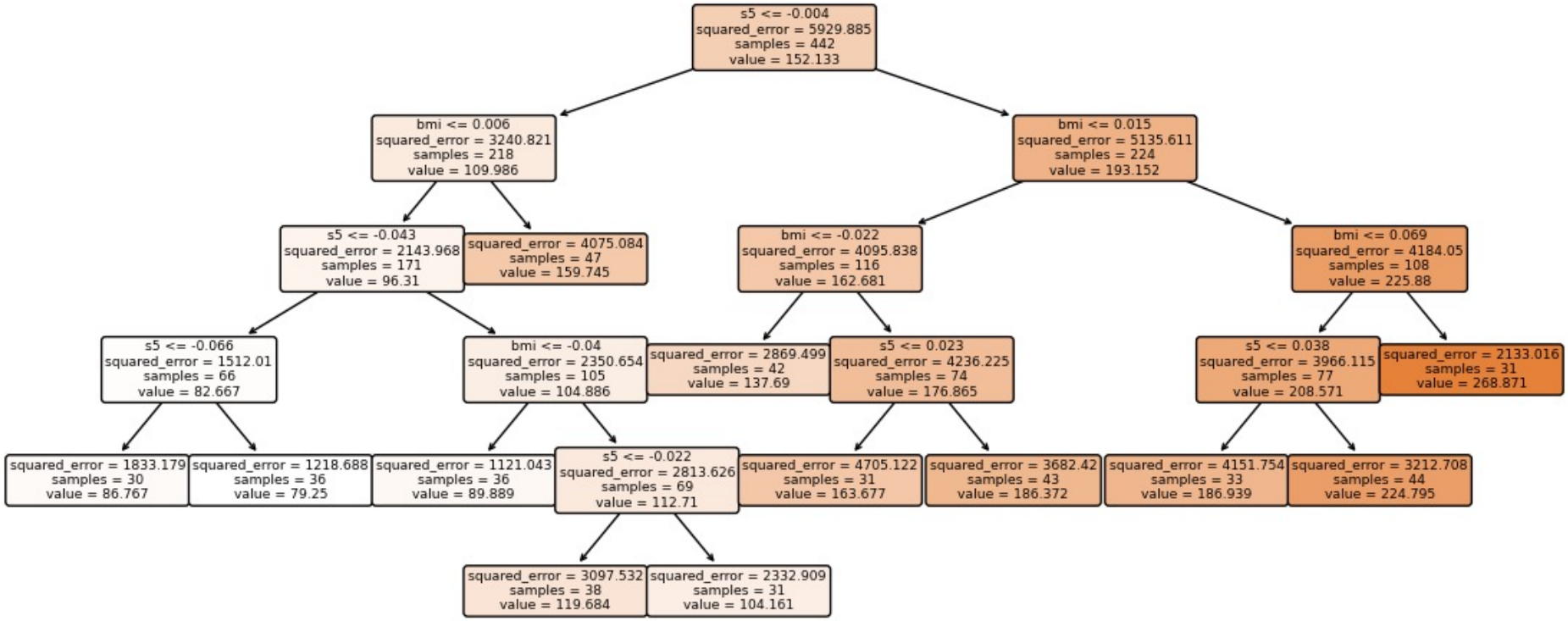
Random basis functions depend on position of sampled input data

No fixed size parameter vector to optimise

Model is called "non-parametric"



Famous Non-parametric Model: Trees

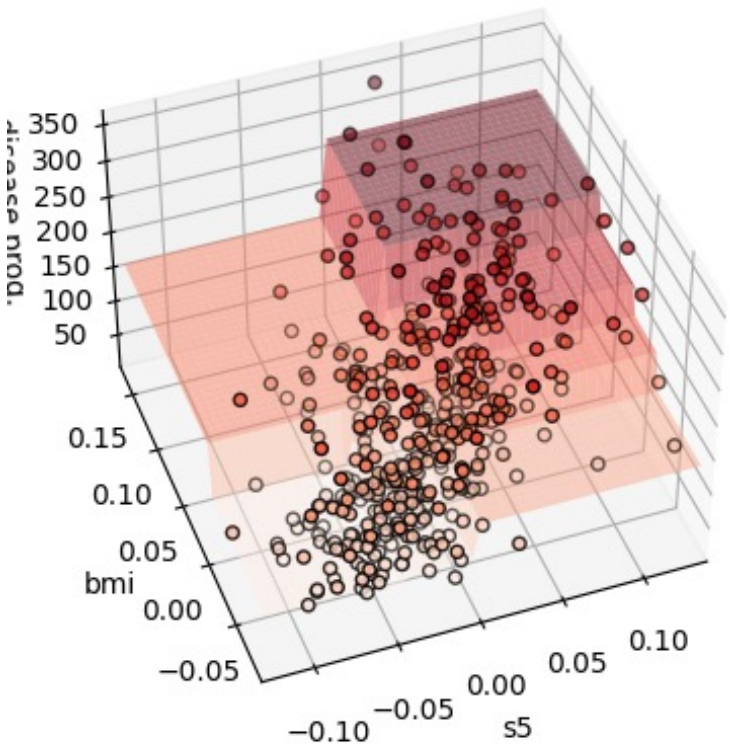


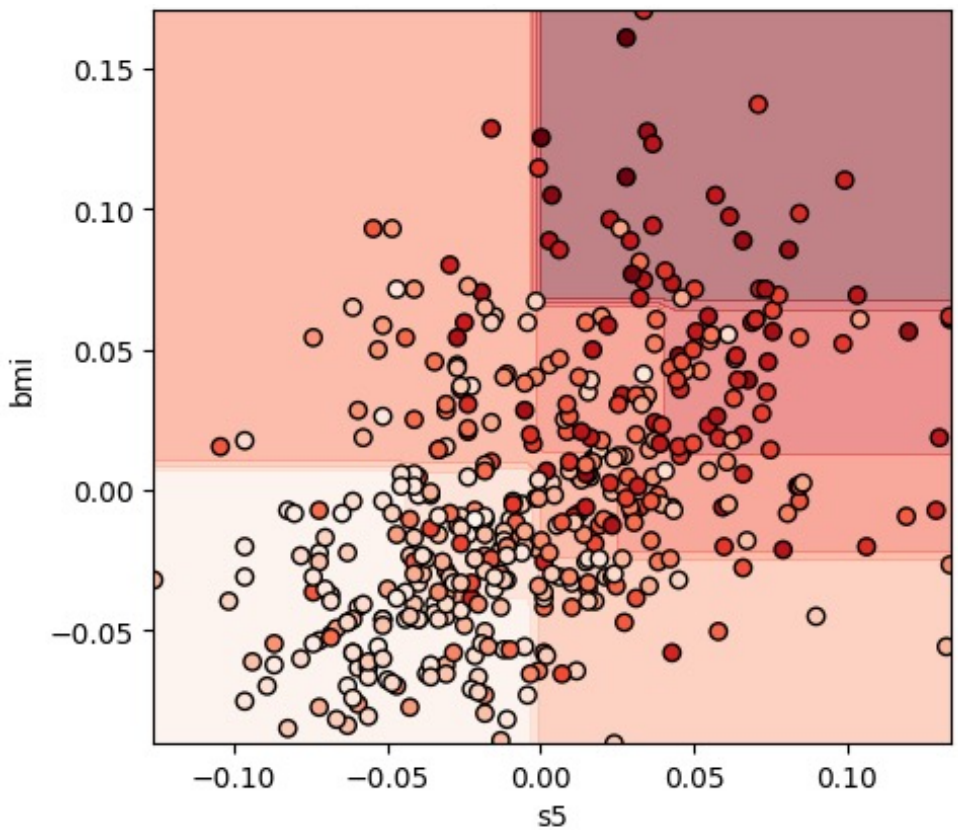
Classification (decision) and regression trees:

- Applying sequence of tests to datapoint $\boldsymbol{x}$ to find matching leaf node $l(\boldsymbol{x})$
- Provide one constant prediction per leaf $y(\boldsymbol{x}, \boldsymbol{w}, T) = w_{l(\boldsymbol{x})}$

Growing tree structure T and number of leaves growing N

Famous Non-parametric Model: Trees





Like KNN piecewise constant prediction function, but with axis-parallel regions.

Parametric Models – Increasing *N*

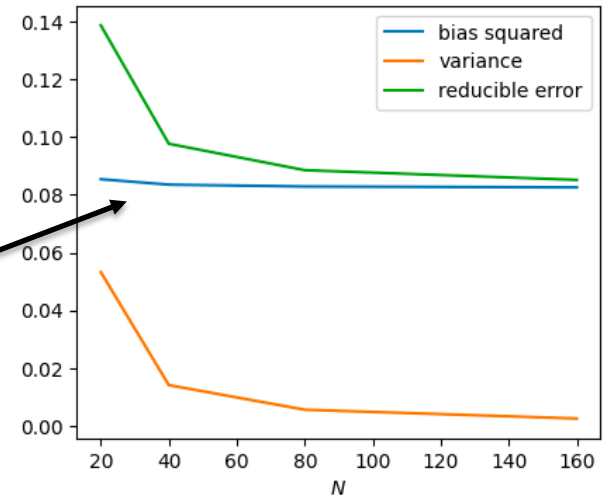
$$\mathbb{E}\big(h(X) - y(X)\big)^2$$

reducible error

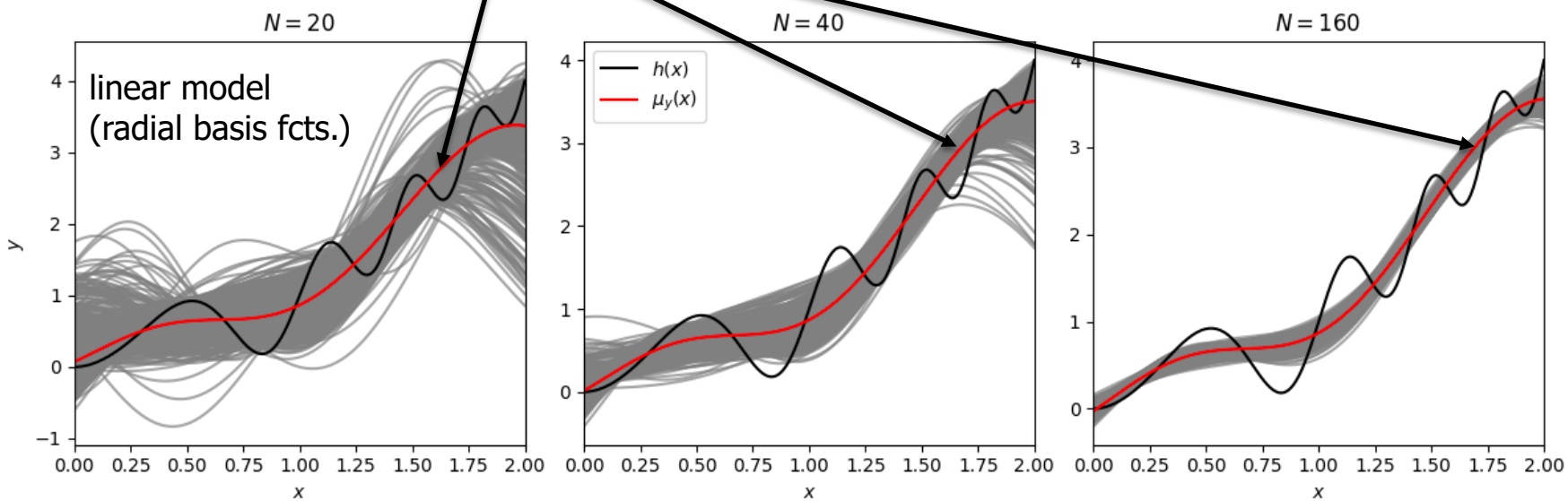
$$= \mathbb{E}\left(h(X) - \mu_y(X)\right)^2 + \mathbb{E}\left(y(X) - \mu_y(X)\right)^2$$

bias square

variance



bias essentially constant



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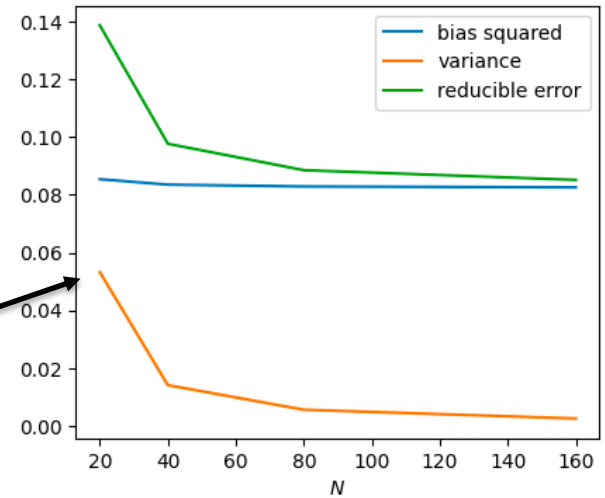
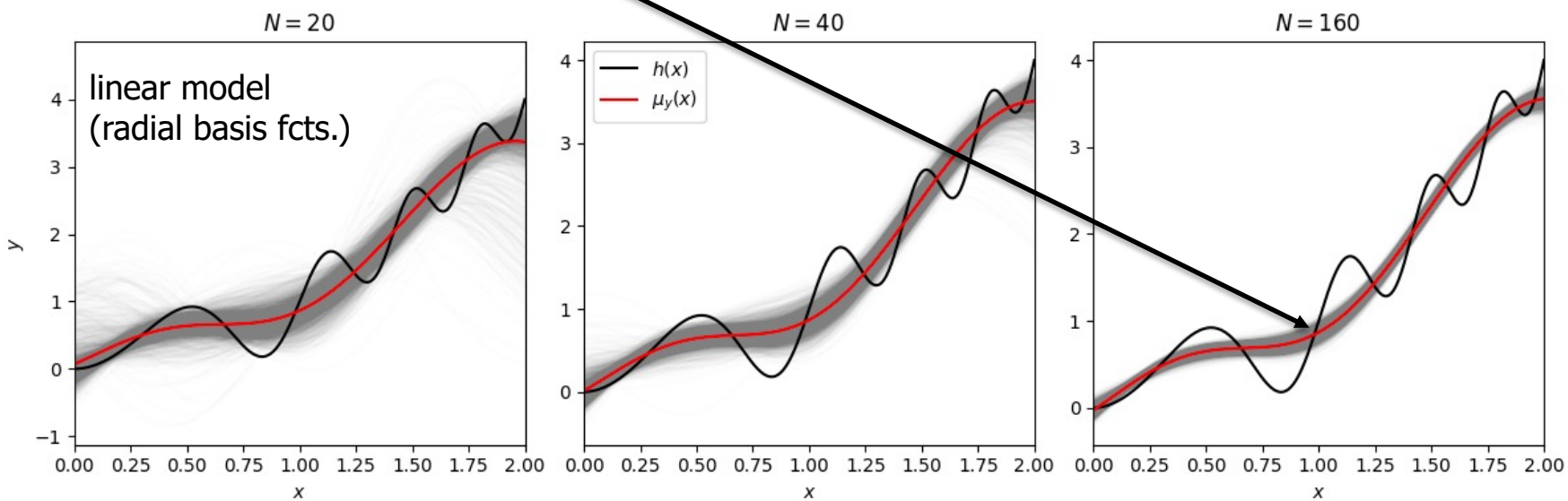


bias essentially constant

variance vanishes



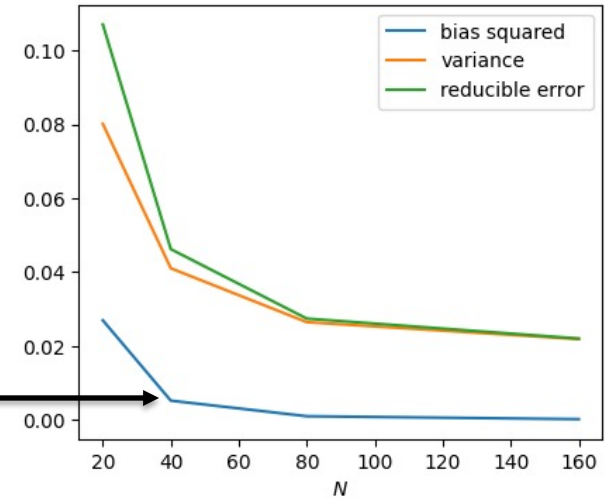
Non-Parametric Models – Increasing *N*

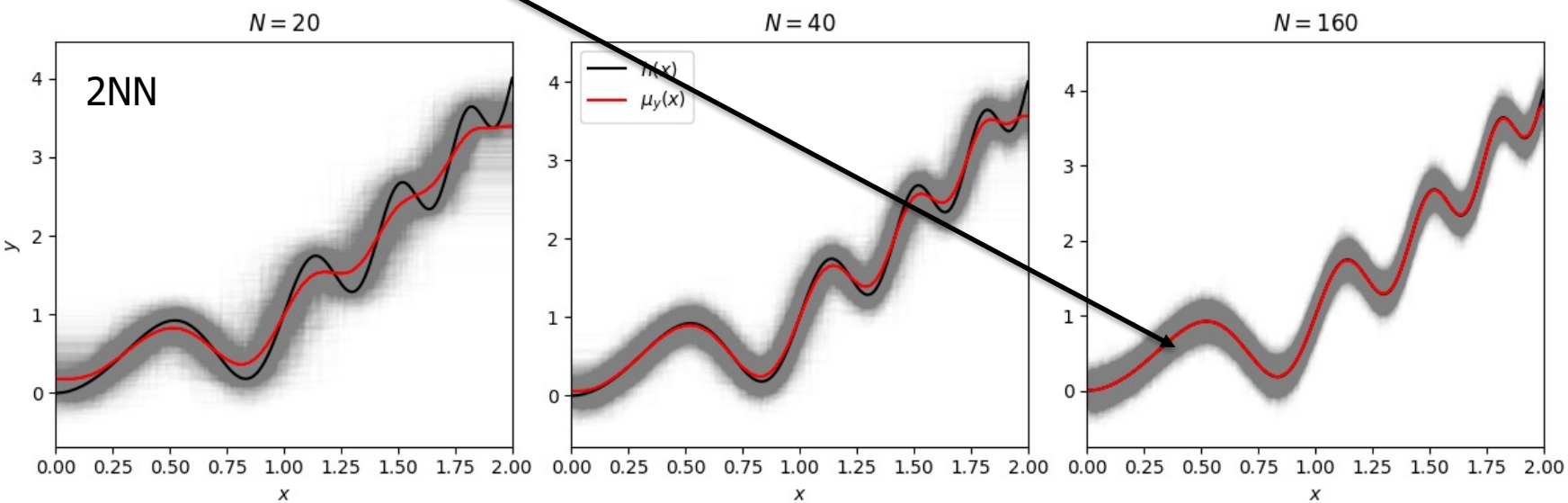
$$\mathbb{E}\big(h(X) - y(X)\big)^2$$ reducible error

$$= \mathbb{E}\left(h(X) - \mu_y(X)\right)^2 + \mathbb{E}\left(y(X) - \mu_y(X)\right)^2$$

bias square variance

bias can vanish



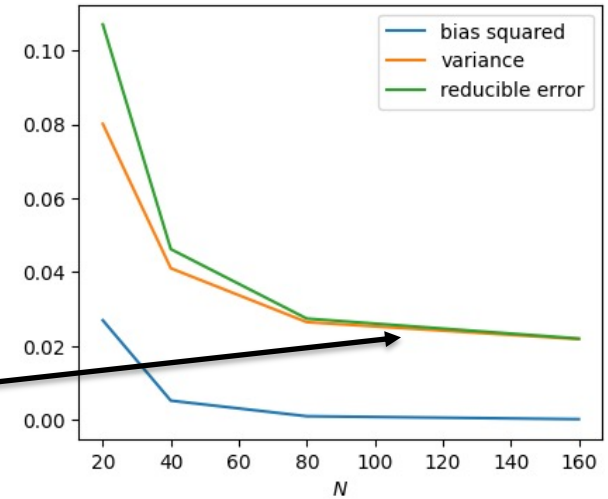


Non-Parametric Models – Increasing *N*

$$\mathbb{E}\big(h(X) - y(X)\big)^2$$ reducible error

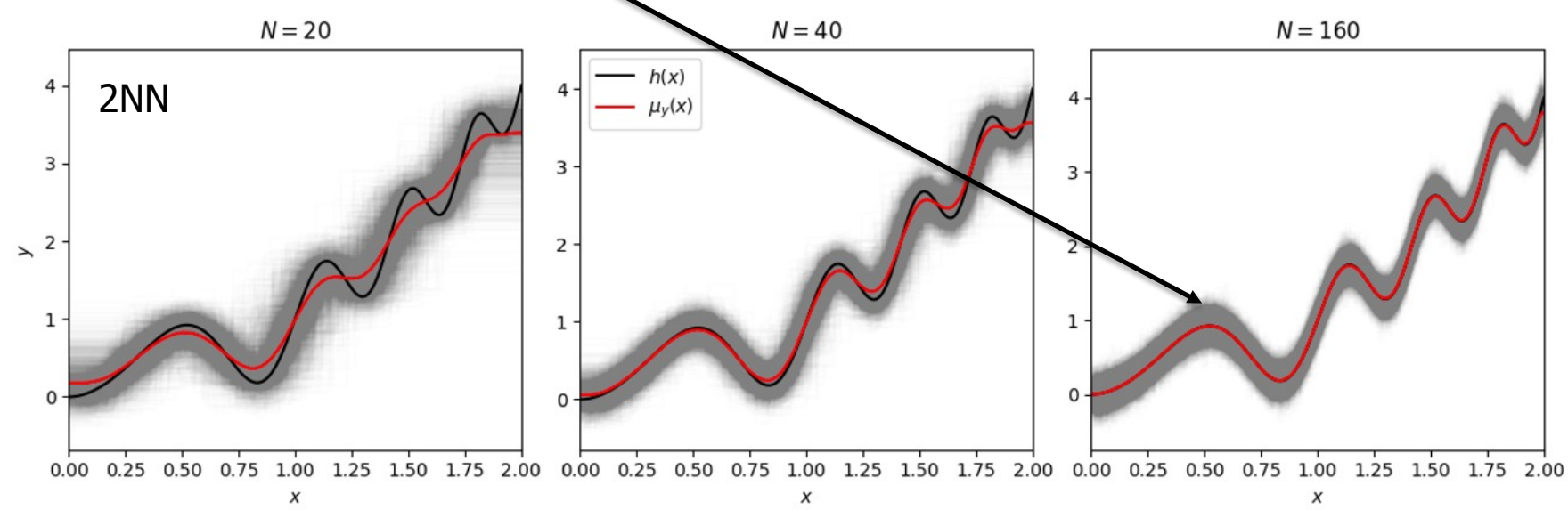
$$= \mathbb{E}\left(h(X) - \mu_y(X)\right)^2 + \mathbb{E}\left(y(X) - \mu_y(X)\right)^2$$

bias square variance



bias can vanish

but variance might not (could even increase)

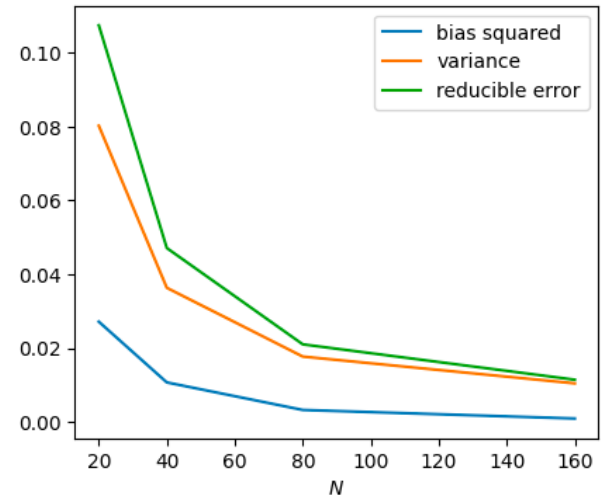


Decrease Variance by Hyper-parameter

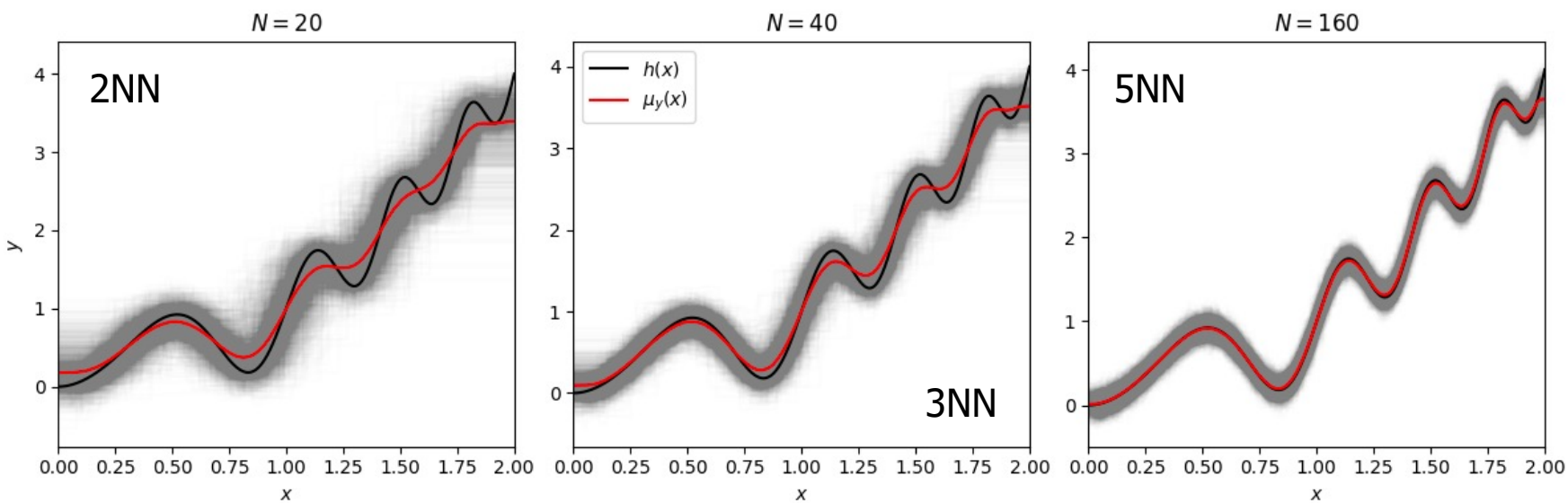
Idea: Choose architectural parameters with lower variance for increasing N

Achieve: $\boxed{\lim_{N\to\infty} \mathbb{E}\big(h(X) - y(X)\big)^2 = 0}$

called "consistency"



Example: KNN consistent if $K \to \infty$ and $K/N \to 0$



Summary: Non-parametric Models

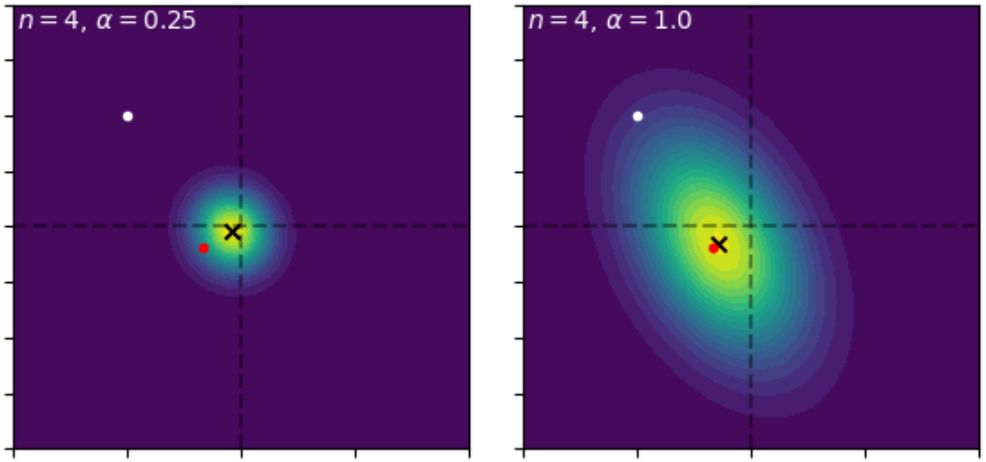
- Non-parametric models cannot be defined by finite set of parameters (independent of N)
- Examples: *k*NN, classification/regression trees and additive combinations of them (forests), Gaussian process regression (later this Module)
- In contrast to parametric models: bias typically vanishes with N, but variance might not vanish
- Can use less flexible models for larger N to achieve consistency (reducible error converges to 0)

Advanced Topics

Bayesian Prediction and Model Selection

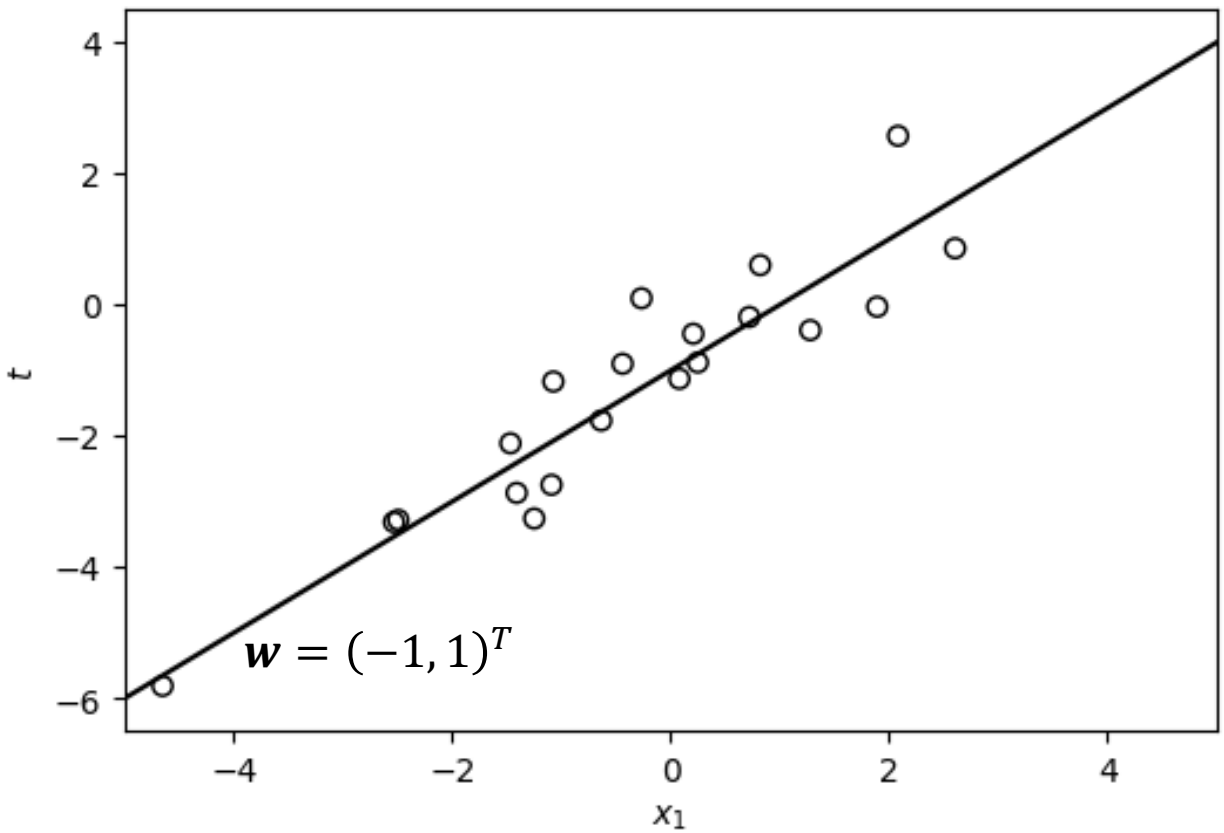
using the example of

Bayesian Ridge Regression



Recall: Ridge Regression

$$\boldsymbol{t} \mid \boldsymbol{X} \sim \mathrm{N}_n(\boldsymbol{X}\boldsymbol{w}, \sigma^2 \boldsymbol{I}),\ \boldsymbol{X} = [\boldsymbol{1}, \boldsymbol{x}_1],\ \boldsymbol{x}_1 \sim \mathrm{N}_n(\boldsymbol{0}, 2\boldsymbol{I})$$



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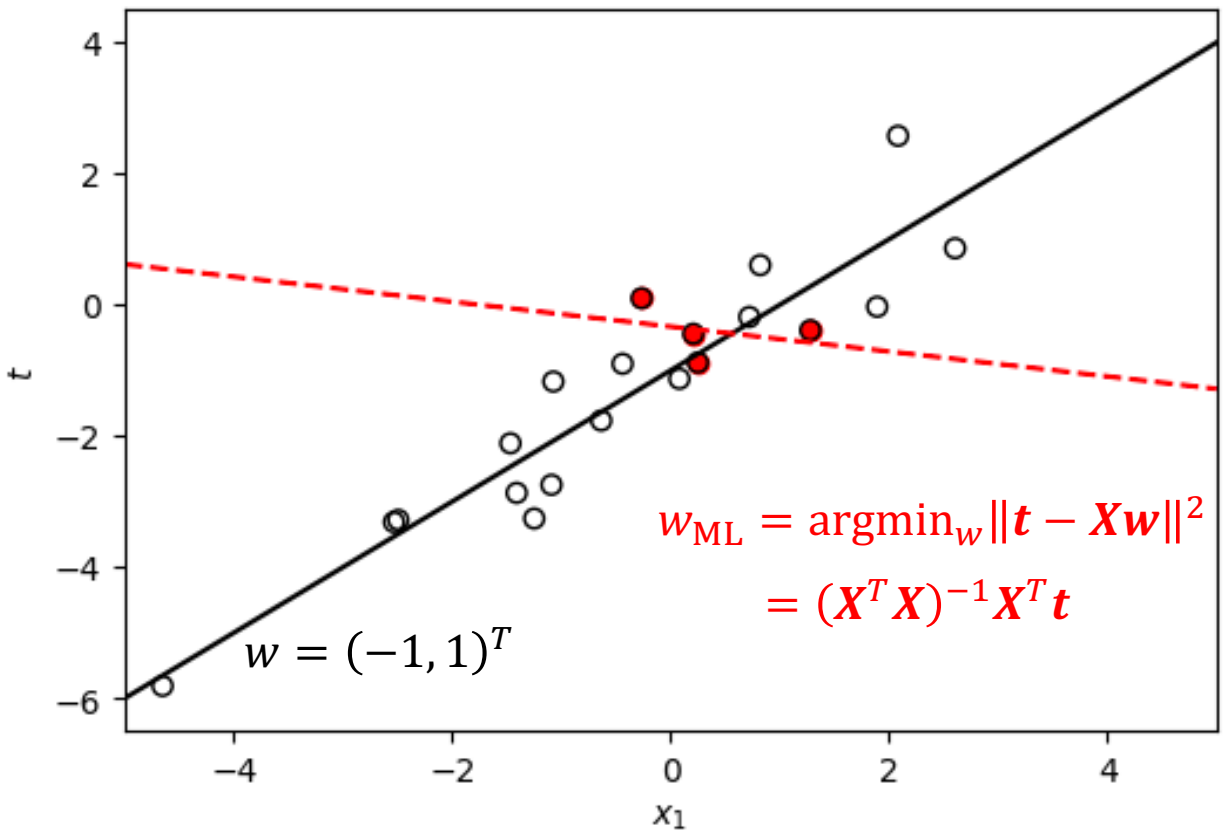
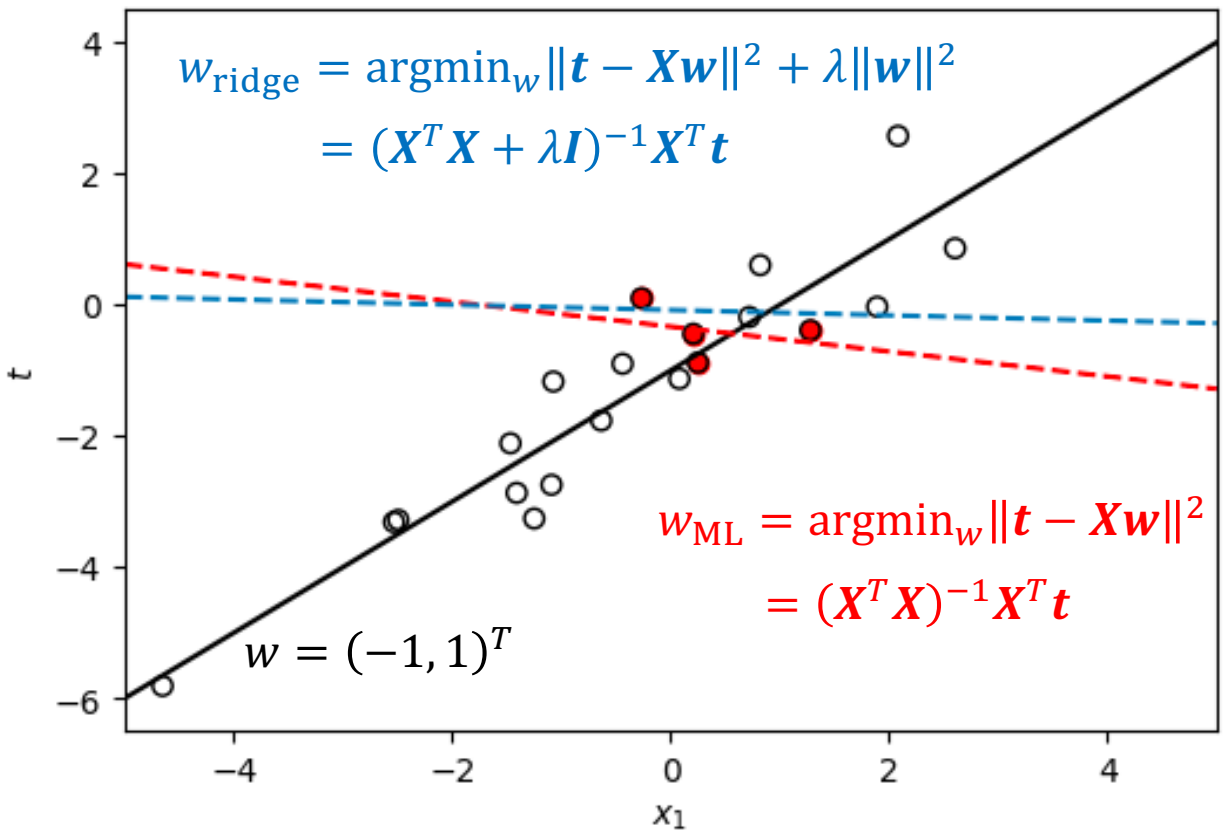


Maximum likelihood estimate minimises sum of squared errors

Recall: Ridge Regression

$$\boldsymbol{t} \mid \boldsymbol{X} \sim N_n(\boldsymbol{X}\boldsymbol{w}, \sigma^2 \boldsymbol{I}),\ \boldsymbol{X} = [\boldsymbol{1}, \boldsymbol{x}_1],\ \boldsymbol{x}_1 \sim N_n(\boldsymbol{0}, 2\boldsymbol{I})$$



Ridge regression objective "shrinks" ML solution.
Useful to avoid overfitting.

Maximum likelihood estimate minimises sum of squared errors

Mentioned in passing: "can derive ridge regression as Bayesian method"

What is Bayesian Approach to ML

1. Define probabilistic model (likelihood) $p(t|\boldsymbol{X}, \boldsymbol{w})$ as before

2. Assume parameters $\boldsymbol{w}$ of model $p(t|\boldsymbol{x}, \boldsymbol{w})$ are itself a random variable

3. Define marginal distribution $p(\boldsymbol{w})$, called **prior distribution**, that captures prior knowledge / belief about parameter

4. Compute **posterior distribution** $p(\boldsymbol{w}|D)$ with Bayes' Theorem

$$p(\boldsymbol{w}|D) = \frac{p(\boldsymbol{t}|\boldsymbol{X}, \boldsymbol{w})p(\boldsymbol{w})}{p(\boldsymbol{t}|\boldsymbol{X})}$$

posterior

output likelihood

prior

This directly gives us uncertainty about parameters!

Bayesian Ridge Regression

What is meant by "deriving ridge regression as Bayesian method"?

We can choose prior (and model/likelihood) such that ridge solution

$$\boldsymbol{w}_{\text{ridge}} = (\boldsymbol{X}^T\boldsymbol{X} + \lambda\boldsymbol{I})^{-1}\boldsymbol{X}^T\boldsymbol{t}$$

corresponds to posterior mean (and mode i.e. maximum posterior MAP)

$$\boldsymbol{w}_{\text{ridge}} = \mathbf{E}(\boldsymbol{w} \mid \boldsymbol{X}, \boldsymbol{t}) = \int_{\mathbb{R}^p} \boldsymbol{w}\, p(\boldsymbol{w} \mid \boldsymbol{X}, \boldsymbol{t})\, \mathrm{d}\boldsymbol{w}$$

How?

$\boldsymbol{t} \sim \mathrm{N}_n(\boldsymbol{X}\boldsymbol{w}, \sigma^2\boldsymbol{I})$ (our usual model that motivated squared error)

$\boldsymbol{w} \sim \mathrm{N}_p(\boldsymbol{0}, \alpha^2\boldsymbol{I})$ (multivariate normal prior on weights)

Let us convince ourselves by finding posterior distribution!

Derivation of posterior (1/2)

$$p(\boldsymbol{w} \mid \boldsymbol{X}, \boldsymbol{t}) \propto p(\boldsymbol{t} \mid \boldsymbol{X}, \boldsymbol{w}) p(\boldsymbol{w})$$

$$\propto \exp\left(-\frac{1}{2\sigma^2}(\boldsymbol{t} - \boldsymbol{X}\boldsymbol{w})^T(\boldsymbol{t} - \boldsymbol{X}\boldsymbol{w})\right) \exp\left(-\frac{1}{2\alpha^2}\boldsymbol{w}^T\boldsymbol{w}\right)$$

$$\propto \exp\left(-\frac{1}{2\sigma^2}\left(\boldsymbol{w}\boldsymbol{X}^T\boldsymbol{X}\boldsymbol{w} - 2\boldsymbol{t}^T\boldsymbol{X}\boldsymbol{w}\right) - \frac{1}{2\alpha^2}\boldsymbol{w}^T\boldsymbol{w}\right)$$

$$= \exp\left(-\frac{1}{2}\left(\boldsymbol{w}^T \underbrace{\left(\frac{1}{\sigma^2}\boldsymbol{X}^T\boldsymbol{X} + \frac{1}{\alpha^2}\boldsymbol{I}\right)}_{\boldsymbol{A}} \boldsymbol{w} - \underbrace{\frac{2}{\sigma^2}\boldsymbol{t}^T\boldsymbol{X}}_{\boldsymbol{b}^T}\boldsymbol{w}\right)\right)$$

useful technique: drop normalisation constant

(figure out later)

So posterior is exponential function of a degree two polynomial in $\boldsymbol{w}$:

$$p(\boldsymbol{w} \mid \boldsymbol{X}, \boldsymbol{t}) \propto \exp(-(\boldsymbol{w}^T\boldsymbol{A}\boldsymbol{w} + \boldsymbol{b}^T\boldsymbol{w})/2)$$

Implies normality with cov. $\boldsymbol{A}^{-1}$, just need to find mean by "completing the square"

$$\boldsymbol{w}^T\boldsymbol{A}\boldsymbol{w} + \boldsymbol{b}^T\boldsymbol{w} = (\boldsymbol{w} - \bar{\boldsymbol{w}})^T\boldsymbol{A}(\boldsymbol{w} - \bar{\boldsymbol{w}}) + c$$

Derivation of posterior (2/2)

Completing the square: for symmetric $\boldsymbol{A} \in \mathbb{R}^{p \times p}$ and $\boldsymbol{b} \in \mathbb{R}^{p}$ we have

$$\boldsymbol{w}^T \boldsymbol{A} \boldsymbol{w} + \boldsymbol{b}^T \boldsymbol{w} = (\boldsymbol{w} - \bar{\boldsymbol{w}})^T \boldsymbol{A} (\boldsymbol{w} - \bar{\boldsymbol{w}}) + c$$

where $\bar{\boldsymbol{w}} = -\boldsymbol{A}^{-1}\boldsymbol{b}/2$ and $c = \bar{\boldsymbol{w}}^T \boldsymbol{A} \bar{\boldsymbol{w}}$.

$$p(\boldsymbol{w} \mid \boldsymbol{X}, \boldsymbol{t}) \propto \exp(-(\boldsymbol{w}^T \boldsymbol{A} \boldsymbol{w} + \boldsymbol{b}^T \boldsymbol{w})/2) \propto \exp(-(\boldsymbol{w} - \bar{\boldsymbol{w}})^T \boldsymbol{A} (\boldsymbol{w} - \bar{\boldsymbol{w}})/2)$$

Here

$$\boldsymbol{b} = -2\sigma^{-2}\boldsymbol{X}^T\boldsymbol{t}$$

$$\boldsymbol{A} = \sigma^{-2}\boldsymbol{X}^T\boldsymbol{X} + \alpha^{-2}\boldsymbol{I} = \sigma^{-2}(\boldsymbol{X}^T\boldsymbol{X} + \sigma^2\alpha^{-2}\boldsymbol{I})$$

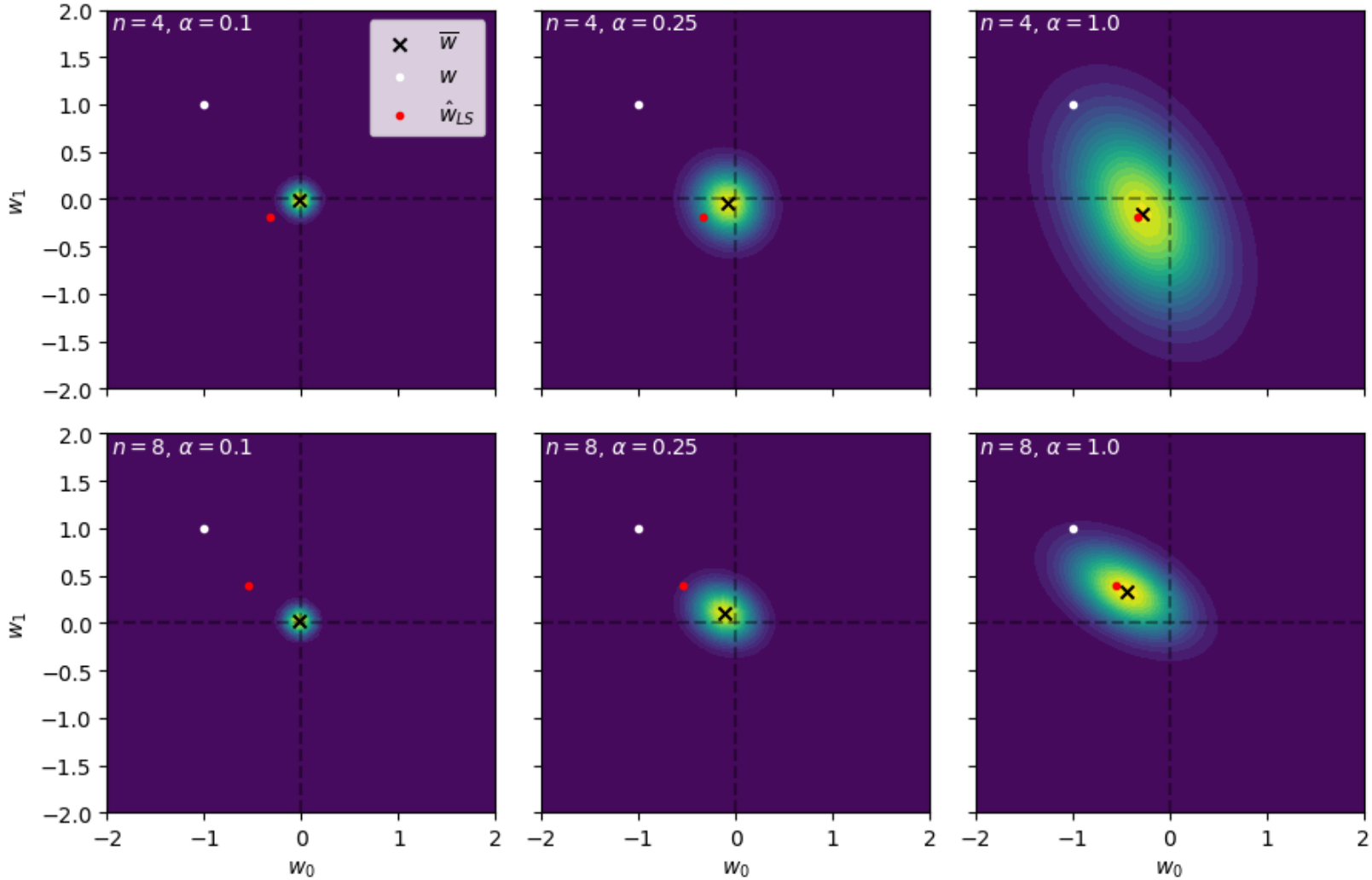
$$\bar{\boldsymbol{w}} = \left(\boldsymbol{X}^T\boldsymbol{X} + \sigma^2\alpha^{-2}\boldsymbol{I}\right)^{-1}\boldsymbol{X}^T\boldsymbol{t}$$

unnormalised version of normal pdf with mean $\bar{\boldsymbol{w}}$ and covariance $\boldsymbol{A}^{-1}$

Thus, posterior distribution: $\boldsymbol{w} \mid \boldsymbol{X}, \boldsymbol{t} \sim \mathrm{N}_p(\bar{\boldsymbol{w}}, \boldsymbol{A}^{-1})$

Posterior Evolution for Example

$$\boldsymbol{w} \mid \boldsymbol{X}, \boldsymbol{t} \sim \mathrm{N}_p(\overline{\boldsymbol{w}}, \boldsymbol{A}^{-1})$$



Intermediate Summary

The Bayesian model

$$\boldsymbol{t} \mid \boldsymbol{w}, \boldsymbol{X} \sim \mathrm{N}_n(X\boldsymbol{w}, \sigma^2 \boldsymbol{I})$$
$$\boldsymbol{w} \sim \mathrm{N}_p(0, \alpha^2 \boldsymbol{I})$$

yields posterior

$$\boldsymbol{w} \mid \boldsymbol{X}, \boldsymbol{t} \sim \mathrm{N}_p(\bar{\boldsymbol{w}}, \boldsymbol{C})$$

with mean and covariance

$$\bar{\boldsymbol{w}} = \left(\boldsymbol{X}^{\boldsymbol{T}}\boldsymbol{X} + \sigma^2\alpha^{-2}\boldsymbol{I}\right)^{-1}\boldsymbol{X}^{\boldsymbol{T}}\boldsymbol{t}$$
$$\boldsymbol{C} = \left(\sigma^{-2}\boldsymbol{X}^{\boldsymbol{T}}\boldsymbol{X} + \alpha^{-2}\boldsymbol{I}\right)^{-1}$$

where mean equal to ridge regression solution (where $\lambda = \sigma^2\alpha^{-2}$)

Questions

- ❑ How to use model for prediction?
- ❑ How to choose model (i.e., α and σ)?

Question 1: How to Predict?

Old answer: simple use predict target value of test point x_* via

$$y(\boldsymbol{x}_*, \overline{\boldsymbol{w}}) = \boldsymbol{x}_*^T \overline{\boldsymbol{w}}$$

Issues:

- No uncertainty quantification
- Throws away all information of posterior other than mean

New answer: Use predictive distribution from posterior model averaging

$$p(t_* \mid \boldsymbol{x}_*, \boldsymbol{X}, \boldsymbol{t}) = \int_{\mathbb{R}^p} p(t_* \mid \boldsymbol{x}_*, \boldsymbol{w}) p(\boldsymbol{w} \mid \boldsymbol{X}, \boldsymbol{t}) \, \mathrm{d}\boldsymbol{w}$$

- Provides prediction $y(\boldsymbol{x}_*) = \mathbb{E}(T \mid X = x_*) = \int_{-\infty}^{\infty} t \, p(t_* \mid \boldsymbol{x}_*, \boldsymbol{X}, \boldsymbol{t}) \, \mathrm{d}t$
- … well as uncertainty about prediction
- Both take uncertainty about weights into account

Predictive Distribution

How to compute predictive distribution

$$p(t_* \mid \boldsymbol{x}_*, \boldsymbol{X}, \boldsymbol{t}) = \int_{\mathbb{R}^p} p(t_* \mid \boldsymbol{x}_*, \boldsymbol{w}) p(\boldsymbol{w} \mid \boldsymbol{X}, \boldsymbol{t}) \, \mathrm{d}\boldsymbol{w}$$

Let's make use of normal posterior $\boldsymbol{w} \mid \boldsymbol{X}, \boldsymbol{t} \sim \mathrm{N}_p(\bar{\boldsymbol{w}}, \boldsymbol{C})$

Recall, linear combination $\boldsymbol{a}^T \boldsymbol{z}$ of normal random vector $\boldsymbol{z} \sim \mathrm{N}_d(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is normal:

$$\boldsymbol{a}^T \boldsymbol{z} \sim \mathrm{N}\left(\boldsymbol{a}^T \boldsymbol{\mu}, \boldsymbol{a}^T \boldsymbol{\Sigma} \boldsymbol{a}\right)$$

So posterior-averaged expected target value (w/o noise) is $\boldsymbol{x}_*^T \boldsymbol{w} \sim N\left(\boldsymbol{x}_*^T \bar{\boldsymbol{w}}, \boldsymbol{x}_*^T \boldsymbol{C} \boldsymbol{x}_*\right)$!

By adding noise $t_* = \boldsymbol{x}_*^T \boldsymbol{w} + \epsilon$ with $\epsilon \sim \mathrm{N}(0, \sigma^2)$ find predictive distribution

$$t_* \mid \boldsymbol{x}_*, \boldsymbol{X}, \boldsymbol{t} \sim \mathrm{N}\left(\boldsymbol{x}_*^T \bar{\boldsymbol{w}}, \boldsymbol{x}_*^T \boldsymbol{C} \boldsymbol{x}_* + \sigma^2\right)$$

prediction is as expected

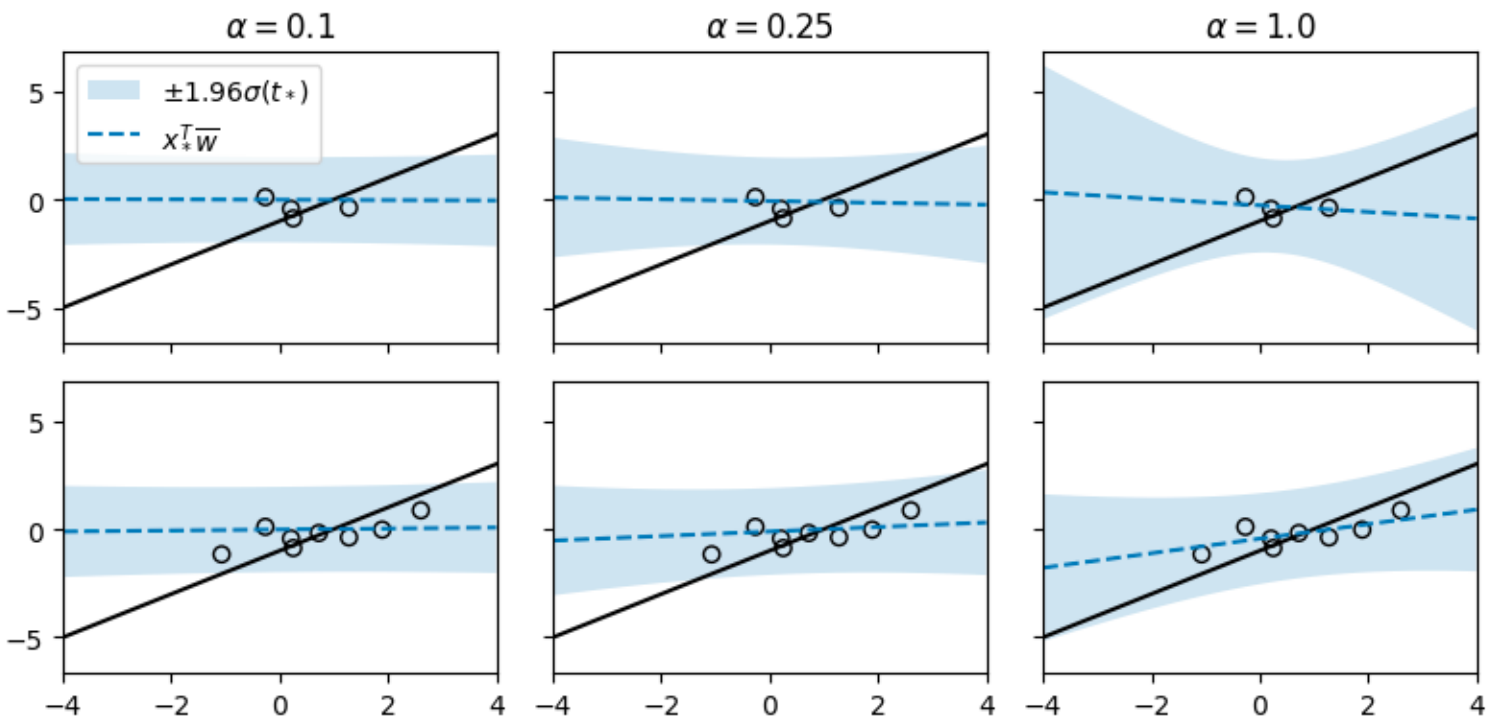
but also get variance to quantify uncertainty (including epistemic uncertainty in first term)

Uncertainty Quantification

Predictive distribution

$$t_* \mid \boldsymbol{x}_*, \boldsymbol{X}, \boldsymbol{t} \sim \mathrm{N}\left(\boldsymbol{x}_*^T \overline{\boldsymbol{w}}, \boldsymbol{x}_*^{\boldsymbol{T}} \boldsymbol{C} \boldsymbol{x}_* + \sigma^2\right)$$

(where $\boldsymbol{C} = \left(\sigma^{-2}\boldsymbol{X}^{\boldsymbol{T}}\boldsymbol{X} + \alpha^{-2}\boldsymbol{I}\right)^{-1}$)



Question 2: How to Choose Model?

Old answer: pick α and σ via K-fold cross validation

$$\alpha_*, \sigma_* = \mathrm{argmax}_{\alpha,\sigma} \frac{1}{K} \sum_{k=1}^{K} \left\| \boldsymbol{t}_k^{\mathrm{test}} - \boldsymbol{X}_k^{\mathrm{test}} \overline{\boldsymbol{w}}_{\alpha,\sigma} \right\|$$

Issues:

- If K large, high computational cost (requires K ridge regression fits)
- If K small, averages over fits that use little data (might pick poor α, σ)

New answer: pick α and σ to maximise marginal likelihood ("model evidence")

$$p(\boldsymbol{t} \mid \boldsymbol{X}, \alpha, \sigma) = \int_{\mathbb{R}^p} p(\boldsymbol{t} \mid \boldsymbol{X}, \boldsymbol{w}, \sigma)\, p(\boldsymbol{w} \mid \alpha)\, \mathrm{d}\boldsymbol{w}$$

- Corresponds to normalisation constant in posterior distribution
- Trades off spread of prior and how well model can fit data
- Potentially fast if we find simple form of $p(\boldsymbol{t} \mid \boldsymbol{X}, \alpha, \sigma)$

How to Compute Marginal Likelihood?

Suppose we only have single (training) data point $(\boldsymbol{x}, t)$

Recall, linear combination $\boldsymbol{a}^{\boldsymbol{T}}\boldsymbol{z}$ of normal random vector $\boldsymbol{z} \sim \mathbf{N}_{\boldsymbol{d}}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is normal:

$$\boldsymbol{a}^{\boldsymbol{T}}\boldsymbol{z} \sim \mathrm{N}\left(\boldsymbol{a}^{\boldsymbol{T}}\boldsymbol{\mu}, \boldsymbol{a}^{T}\boldsymbol{\Sigma}\boldsymbol{a}\right)$$

So for prior $\boldsymbol{w} \sim \mathrm{N}_p(\mathbf{0}, \alpha^2\boldsymbol{I})$ we get $\quad \boldsymbol{x}^{\boldsymbol{T}}\boldsymbol{w} \sim \mathrm{N}\left(\mathbf{0}, \alpha^2\boldsymbol{x}^{\boldsymbol{T}}\boldsymbol{x}\right)$

$$t \sim \mathrm{N}\left(\mathbf{0}, \alpha^2\boldsymbol{x}^{\boldsymbol{T}}\boldsymbol{x} + \sigma^2\right)$$

And can compute log marginal likelihood as

$$\ln p(t \mid \boldsymbol{x}) = \ln\left(\frac{1}{\sqrt{2\pi(\alpha^2\boldsymbol{x}^{\boldsymbol{T}}\boldsymbol{x} + \sigma^2)}}\exp\left(-\frac{t^2}{2(\alpha^2\boldsymbol{x}^{\boldsymbol{T}}\boldsymbol{x} + \sigma^2)}\right)\right)$$

$$= -\frac{1}{2}\ln 2\pi\left(\alpha^2\boldsymbol{x}^{\boldsymbol{T}}\boldsymbol{x} + \sigma^2\right) - \frac{t^2}{2(\alpha^2\boldsymbol{x}^{\boldsymbol{T}}\boldsymbol{x} + \sigma^2)}$$

How to Compute Marginal Likelihood?

General case: input matrix $\boldsymbol{X}$ and target vector $\boldsymbol{t}$

Linear transformation $\boldsymbol{Az}$ of normal random vector $\boldsymbol{z} \sim \mathrm{N}_{\boldsymbol{d}}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is normal:

$$\boldsymbol{Az} \sim \mathrm{N}_k\left(\boldsymbol{A\mu}, \boldsymbol{A\Sigma A}^{\boldsymbol{T}}\right)$$

So for prior $\boldsymbol{w} \sim \mathrm{N}_p(\boldsymbol{0}, \alpha^2\boldsymbol{I})$ we get $\boldsymbol{Xw} \mid \boldsymbol{X} \sim \mathrm{N}_n(\boldsymbol{0}, \boxed{\alpha^2\boldsymbol{XX}^T})$

$$\boldsymbol{t} \mid \boldsymbol{X} \sim \mathrm{N}_n(\boldsymbol{0}, \boldsymbol{K} + \sigma^2\boldsymbol{I})$$

"kernel matrix" $\boldsymbol{K} = \alpha^2\boldsymbol{XX}^T$ of scaled products $\boldsymbol{K}_{ij} = \alpha\boldsymbol{x}_i^T\boldsymbol{x}_j$

And can compute log marginal likelihood as

$$\ln p(\boldsymbol{t} \mid \boldsymbol{X}) = \ln\left(|2\pi(\boldsymbol{K} + \sigma^2\boldsymbol{I})|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}\boldsymbol{t}^T(\boldsymbol{K} + \sigma^2\boldsymbol{I})^{-1}\boldsymbol{t}\right)\right)$$

$$= -\frac{n}{2}\ln 2\pi - \frac{1}{2}\ln|\boldsymbol{K} + \sigma^2\boldsymbol{I}| - \frac{1}{2}\boldsymbol{t}^T(\boldsymbol{K} + \sigma^2\boldsymbol{I})^{-1}\boldsymbol{t}$$

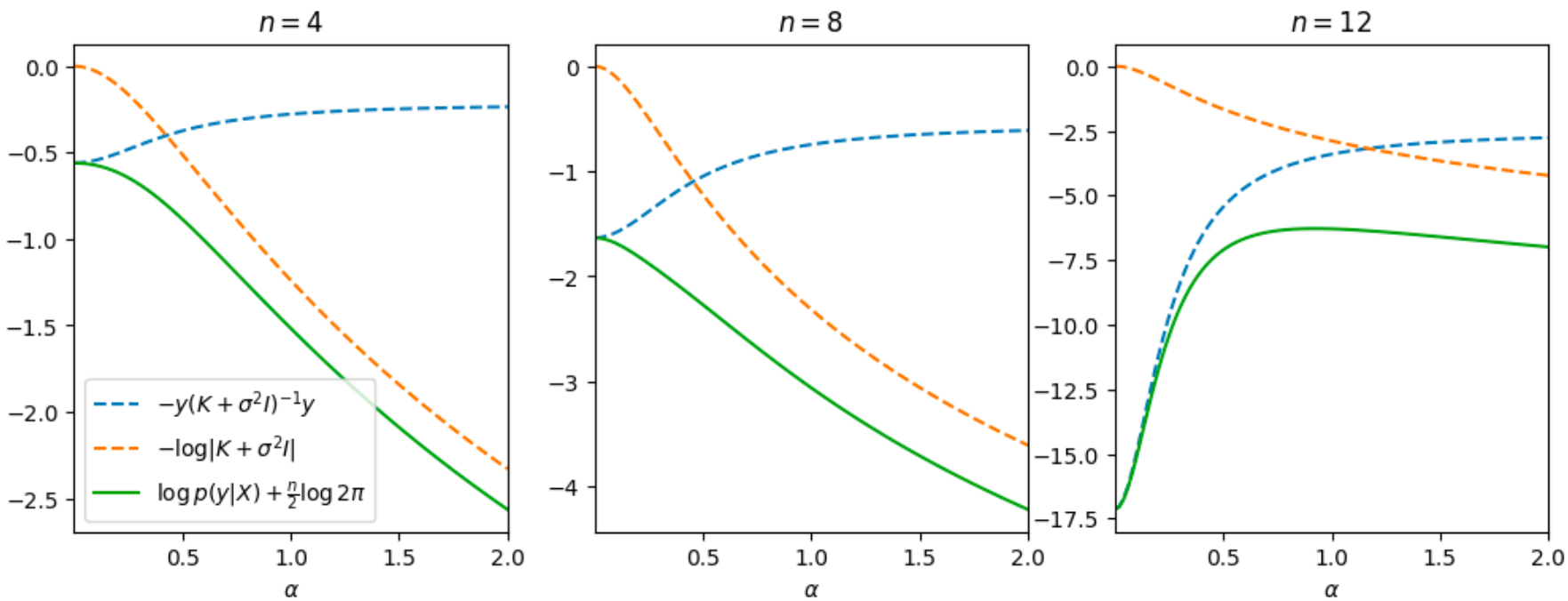
Marginal Likelihood for our Example

$$\ln p(\boldsymbol{t} \mid \boldsymbol{X}, \alpha, \sigma) = -\frac{n}{2}\ln 2\pi - \frac{1}{2}\ln|\boldsymbol{K} + \sigma^2\boldsymbol{I}| - \frac{1}{2}\boldsymbol{t}^T(\boldsymbol{K} + \sigma^2\boldsymbol{I})^{-1}\boldsymbol{t}$$

normalisation

model complexity

model fit



Key Points: Bayesian Ridge Regression

- The Bayesian model with likelihood $\boldsymbol{t} \mid \boldsymbol{w}, \boldsymbol{X} \sim \mathrm{N}_n(X\boldsymbol{w}, \sigma^2)$ and prior $\boldsymbol{w} \sim \mathrm{N}_p(0, \alpha^2)$ yields posterior $\boldsymbol{w} \mid \boldsymbol{X}, \boldsymbol{t} \sim \mathrm{N}_p(\bar{\boldsymbol{w}}, \boldsymbol{C})$ with mean and covariance

$$\bar{\boldsymbol{w}} = \left(\boldsymbol{X}^{\boldsymbol{T}}\boldsymbol{X} + \sigma^2\alpha^{-2}\boldsymbol{I}\right)^{-1}\boldsymbol{X}^{\boldsymbol{T}}\boldsymbol{t}$$
$$\boldsymbol{C} = \left(\sigma^{-2}\boldsymbol{X}^{\boldsymbol{T}}\boldsymbol{X} + \alpha^{-2}\boldsymbol{I}\right)^{-1}$$

- Posterior mean equal to ridge regression solution (where $\lambda = \sigma^2\alpha^{-2}$)
- Provides predictive distribution to quantify predictive uncertainty:

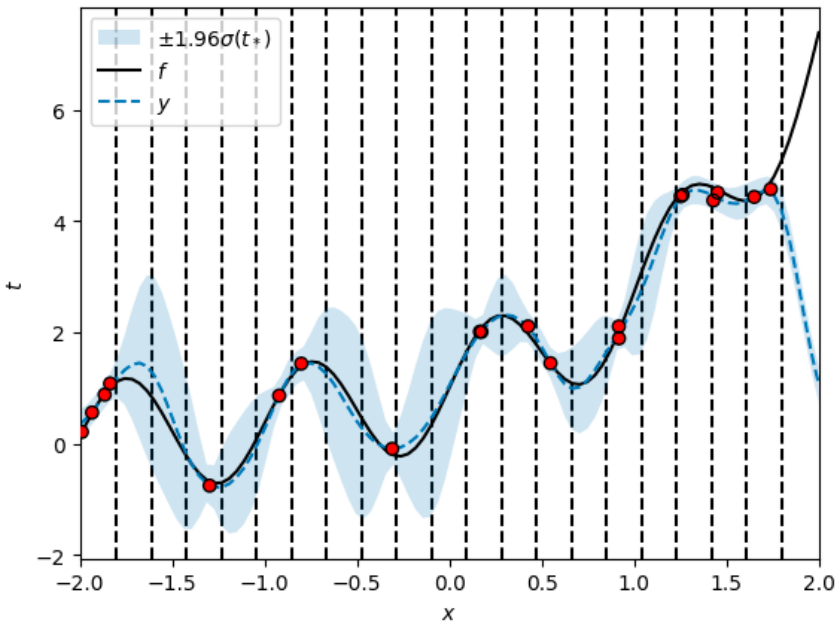
$$t_* \mid \boldsymbol{x}_*, \boldsymbol{X}, \boldsymbol{t} \sim \mathrm{N}\left(\boldsymbol{x}_*^T\bar{\boldsymbol{w}}, \boldsymbol{x}_*^{\boldsymbol{T}}\boldsymbol{C}\boldsymbol{x}_* + \sigma^2\right)$$

- ..and log marginal likelihood for model parameter selection with kernel matrix $\boldsymbol{K} = \alpha^2\boldsymbol{X}\boldsymbol{X}^T$:

$$\ln p(\boldsymbol{t} \mid \boldsymbol{X}, \alpha, \sigma) = -\frac{n}{2}\ln 2\pi - \frac{1}{2}\ln|\boldsymbol{K} + \sigma^2\boldsymbol{I}| - \frac{1}{2}\boldsymbol{t}^T(\boldsymbol{K} + \sigma^2\boldsymbol{I})^{-1}\boldsymbol{t}$$

Advanced Topics

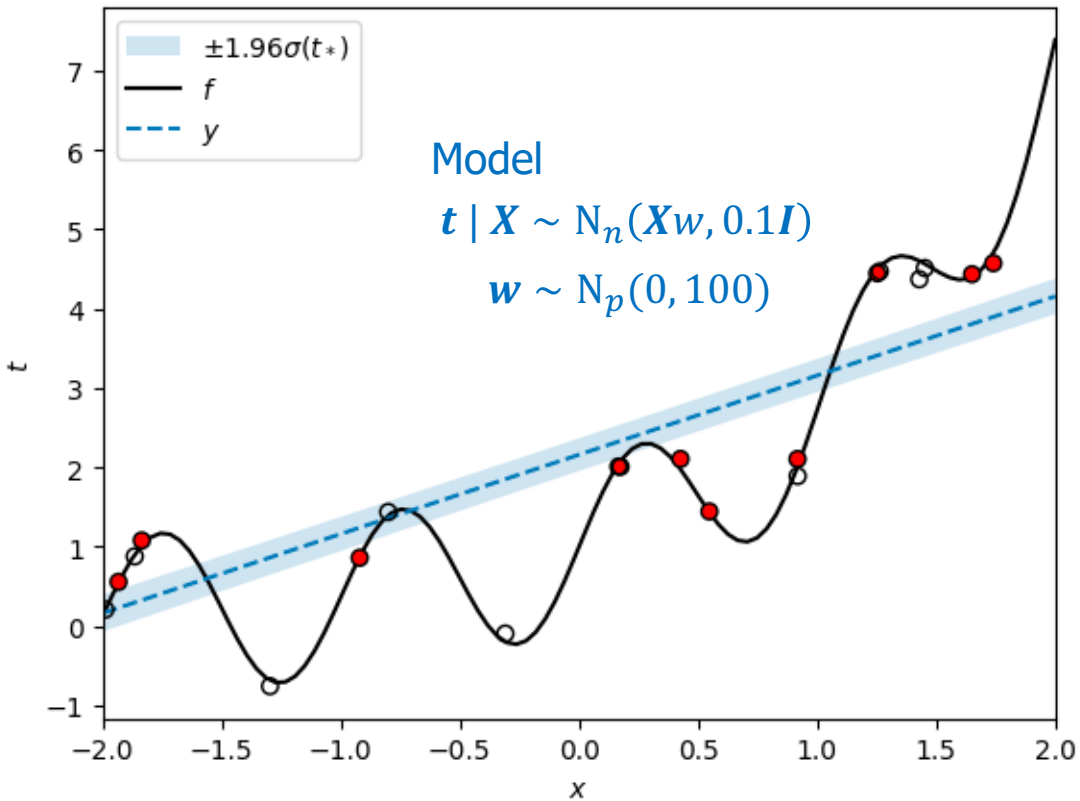
Limits of Parametric Modeling



Non-linear regression function

Data distribution $f(x) = \sin(2\pi x) + \exp(x)$

$$\boldsymbol{t} \mid \boldsymbol{X} \sim \mathrm{N}_n(f(\boldsymbol{X}), 0.1\boldsymbol{I})$$



What goes wrong here?

Misspecified model:

- High bias
- We are very sure about (optimal) weights even with rather wide prior!
- Uncertainty of predictive distribution unrelated to probability of target values

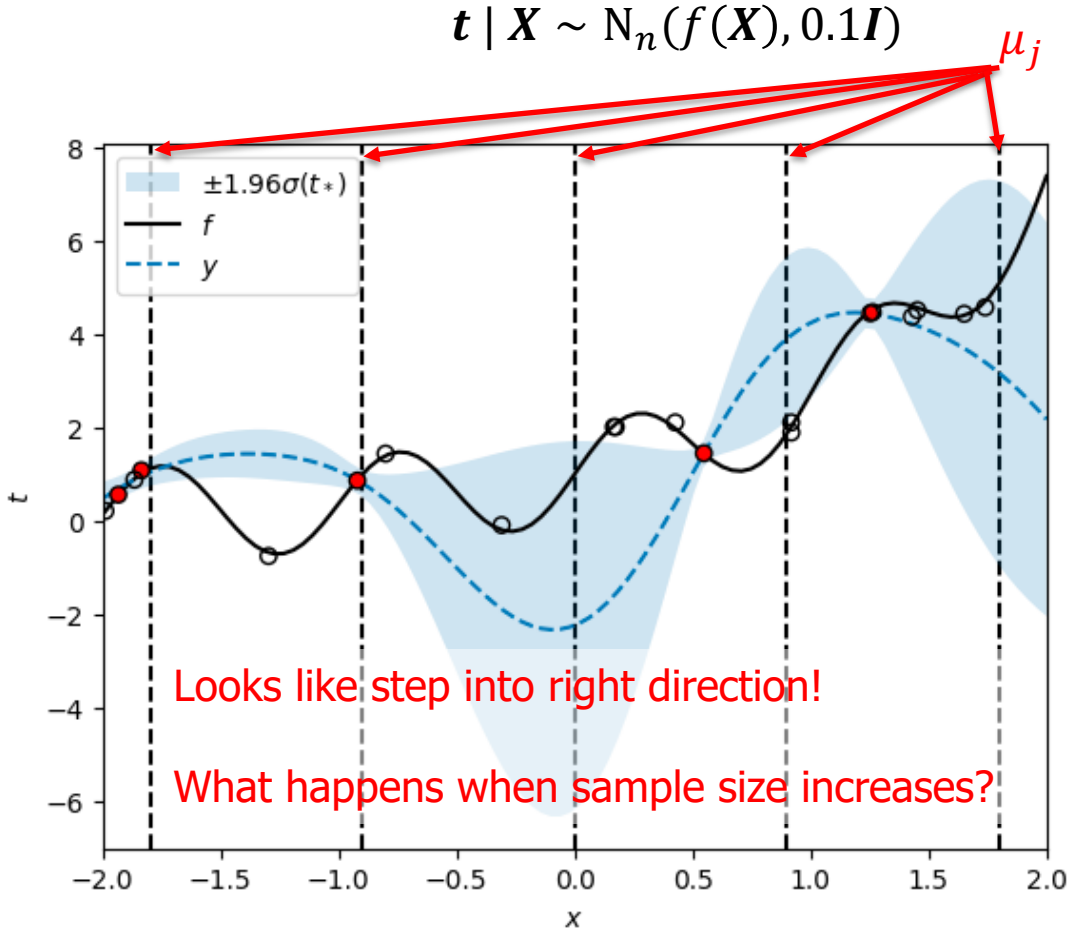
Could try to address this by adapting σ.

…or try lower bias model

Non-linear basis functions

Data distribution $f(x) = \sin(2\pi x) + \exp(x)$

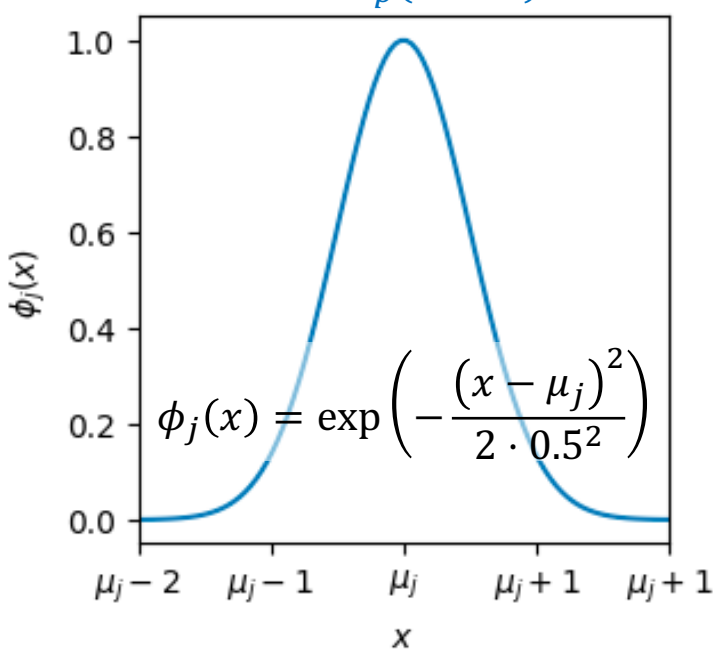
$$\boldsymbol{t} \mid \boldsymbol{X} \sim \mathrm{N}_n(f(\boldsymbol{X}), 0.1\boldsymbol{I})$$



Radial basis functions model

$$\boldsymbol{t} \mid \boldsymbol{X} \sim \mathrm{N}_n(\boldsymbol{\Phi w}, 0.1\boldsymbol{I})$$

$$\boldsymbol{w} \sim \mathrm{N}_p(0, 100)$$



Non-linear basis functions

Data distribution $f(x) = \sin(2\pi x) + \exp(x)$

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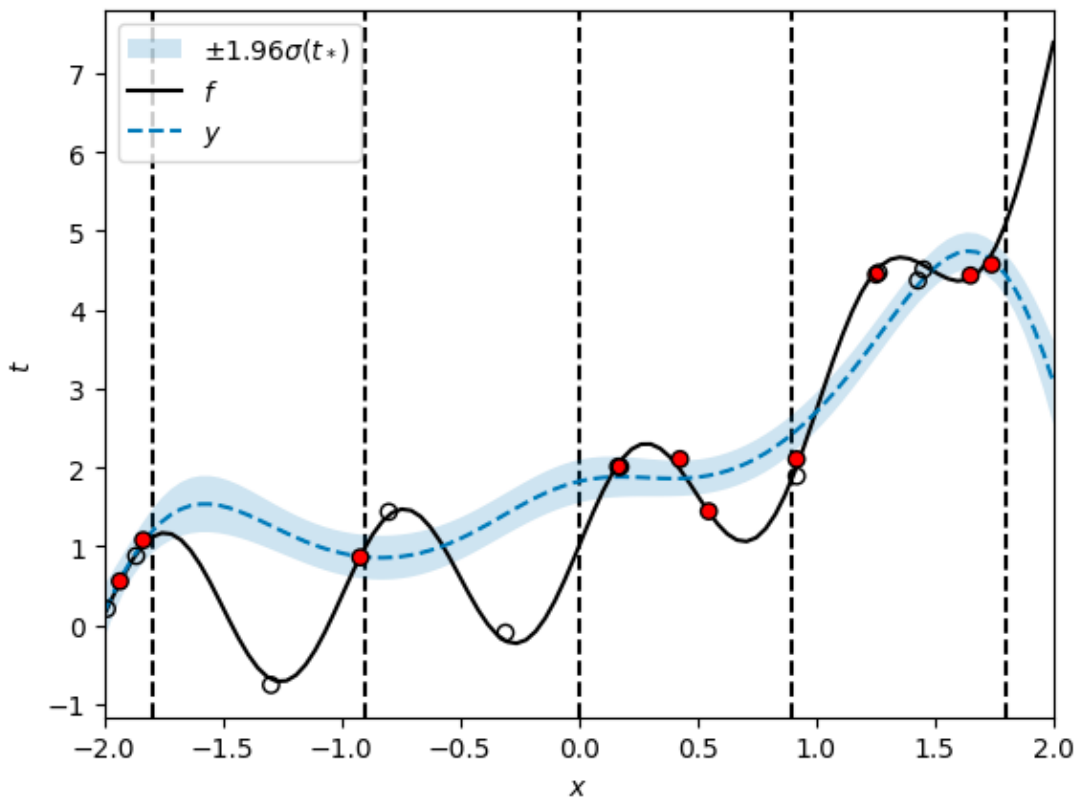

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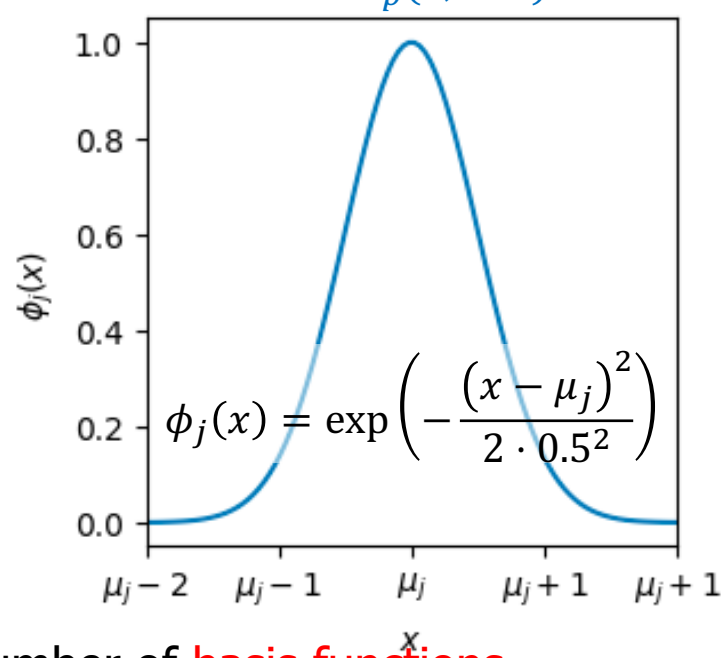
$$\boldsymbol{w} \sim \mathrm{N}_p(0, 100)$$

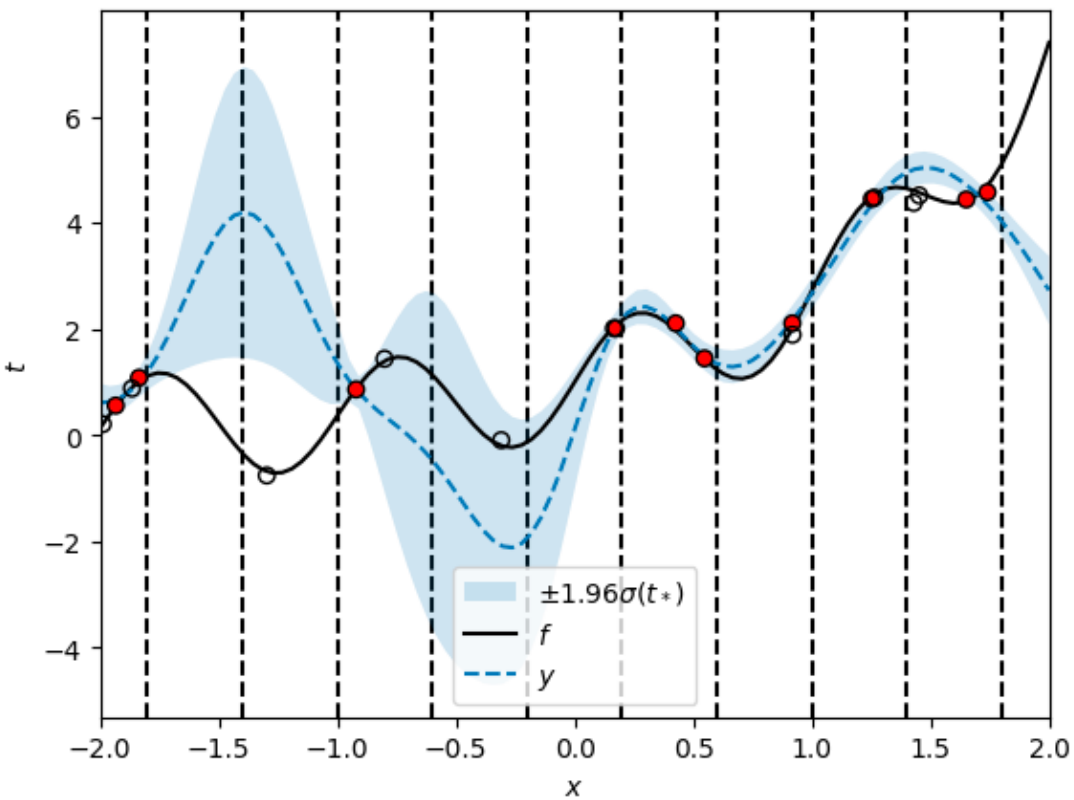


Becoming overconfident again! Try increase number of basis functions.

Non-linear basis functions

Data distribution $f(x) = \sin(2\pi x) + \exp(x)$

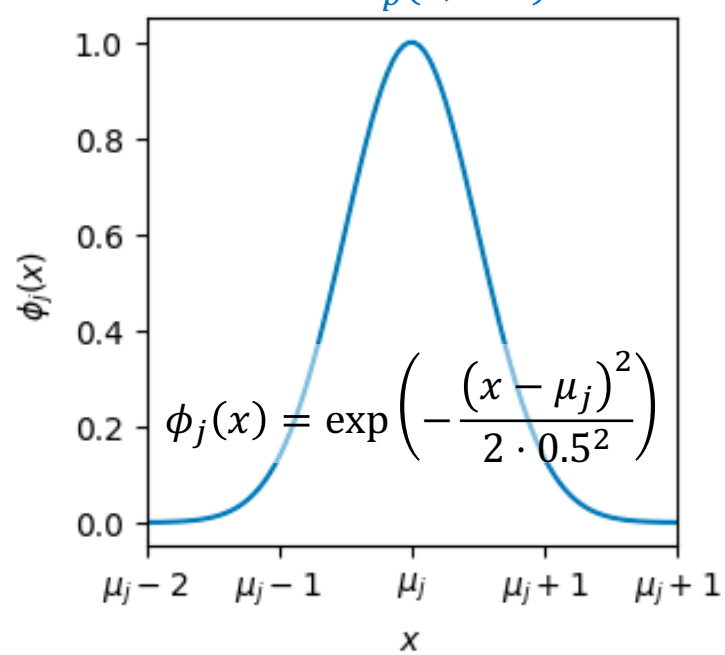
$$\boldsymbol{t} \mid \boldsymbol{X} \sim \mathrm{N}_n(f(\boldsymbol{X}), 0.1\boldsymbol{I})$$



Radial basis functions model

$$\boldsymbol{t} \mid \boldsymbol{X} \sim \mathrm{N}_n(\boldsymbol{\Phi w}, 0.1\boldsymbol{I})$$

$$\boldsymbol{w} \sim \mathrm{N}_p(0, 100)$$

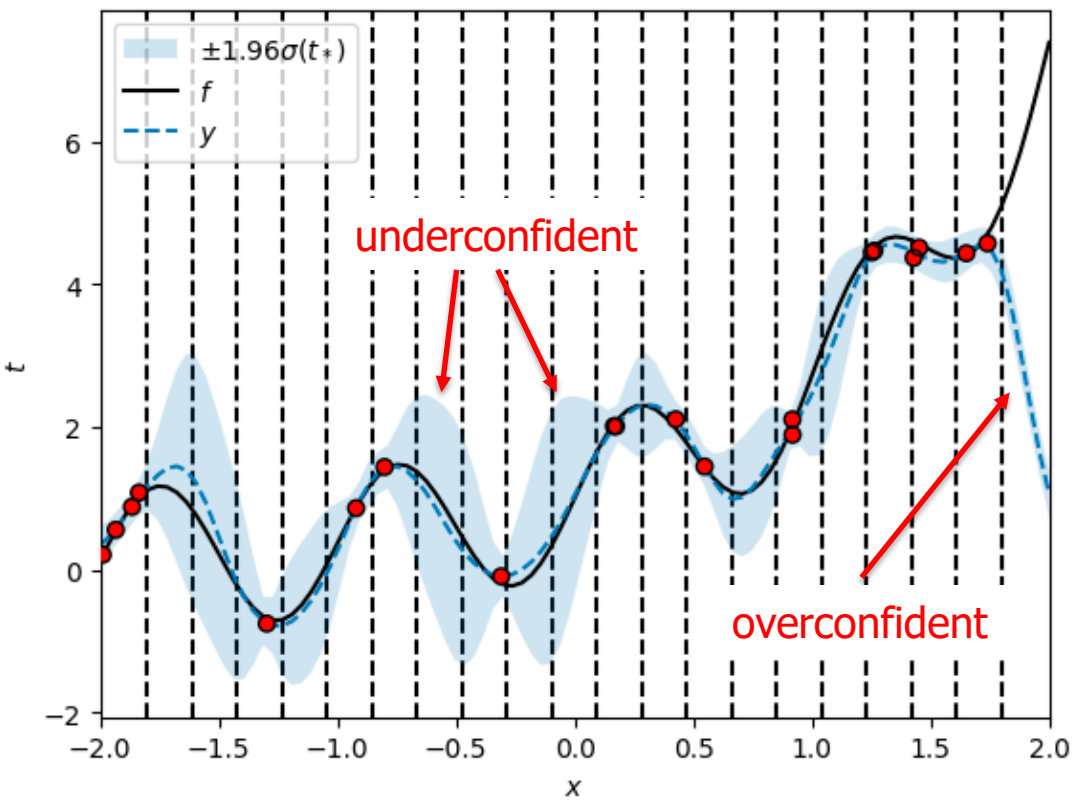


Does this converge to what we want?

Non-linear basis functions

Data distribution $f(x) = \sin(2\pi x) + \exp(x)$

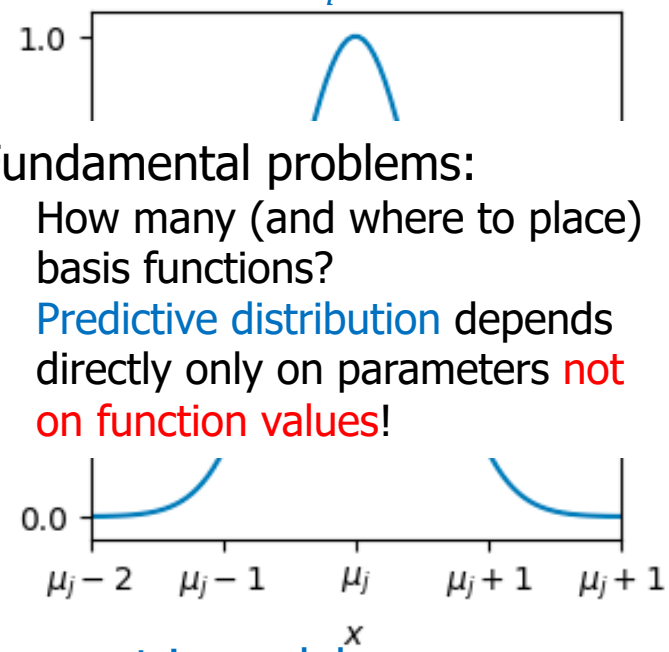
$$\boldsymbol{t} \mid \boldsymbol{X} \sim \mathrm{N}_n(f(\boldsymbol{X}), 0.1\boldsymbol{I})$$



Radial basis functions model

$$\boldsymbol{t} \mid \boldsymbol{X} \sim \mathrm{N}_n(\boldsymbol{\Phi w}, 0.1\boldsymbol{I})$$

$$\boldsymbol{w} \sim \mathrm{N}_p(0, 100)$$



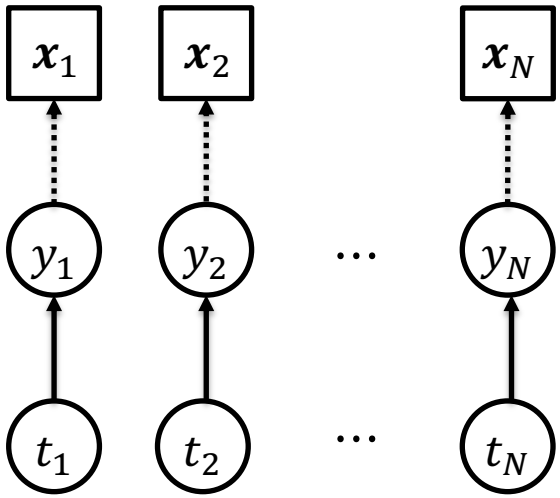
Fundamental problems:

- How many (and where to place) basis functions?
- Predictive distribution depends directly only on parameters not on function values!

Idea: one parameter per function value -> non-parametric model

Advanced Topics

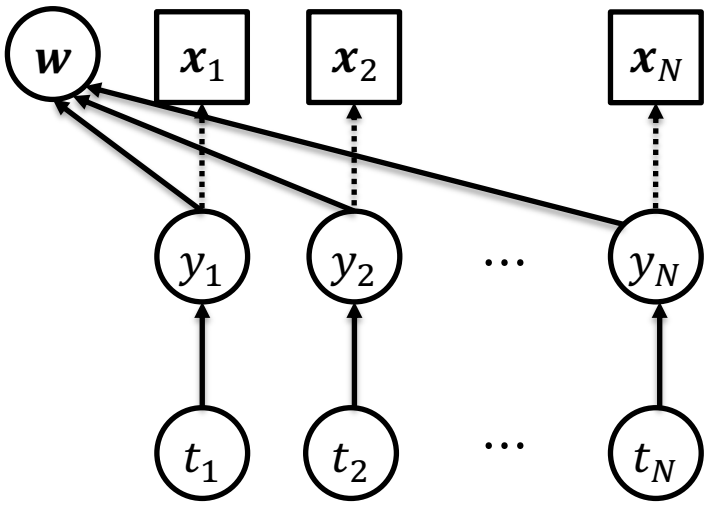
Gaussian Process Regression



From Parametric to Non-parametric

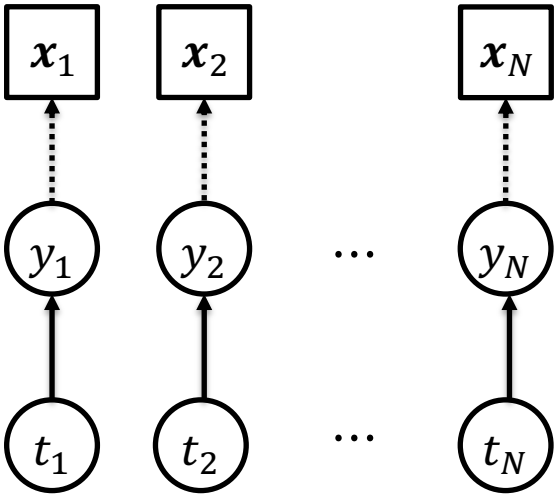
So far:
p-dimensional parameter vector
$\boldsymbol{w} \sim \mathrm{N}_p(\boldsymbol{0}, \alpha^2 \boldsymbol{I})$

Now:
directly use function values
y_n as parameters



$$\boldsymbol{y} \mid \boldsymbol{X} \sim \mathrm{N}_N(\boldsymbol{0}, \boldsymbol{K})$$

$$\boldsymbol{t} \mid \boldsymbol{y}, \boldsymbol{X} \sim \mathrm{N}_N(\boldsymbol{y}, \sigma^2 \boldsymbol{I})$$



$$\boldsymbol{K}_{ij} = \alpha^2 \boldsymbol{x}_i^T \boldsymbol{x}_j$$

covariance structure
of function values $\boldsymbol{y}$

$$\boldsymbol{K}_{ij} = k(\boldsymbol{x}_i, \boldsymbol{x}_j)$$

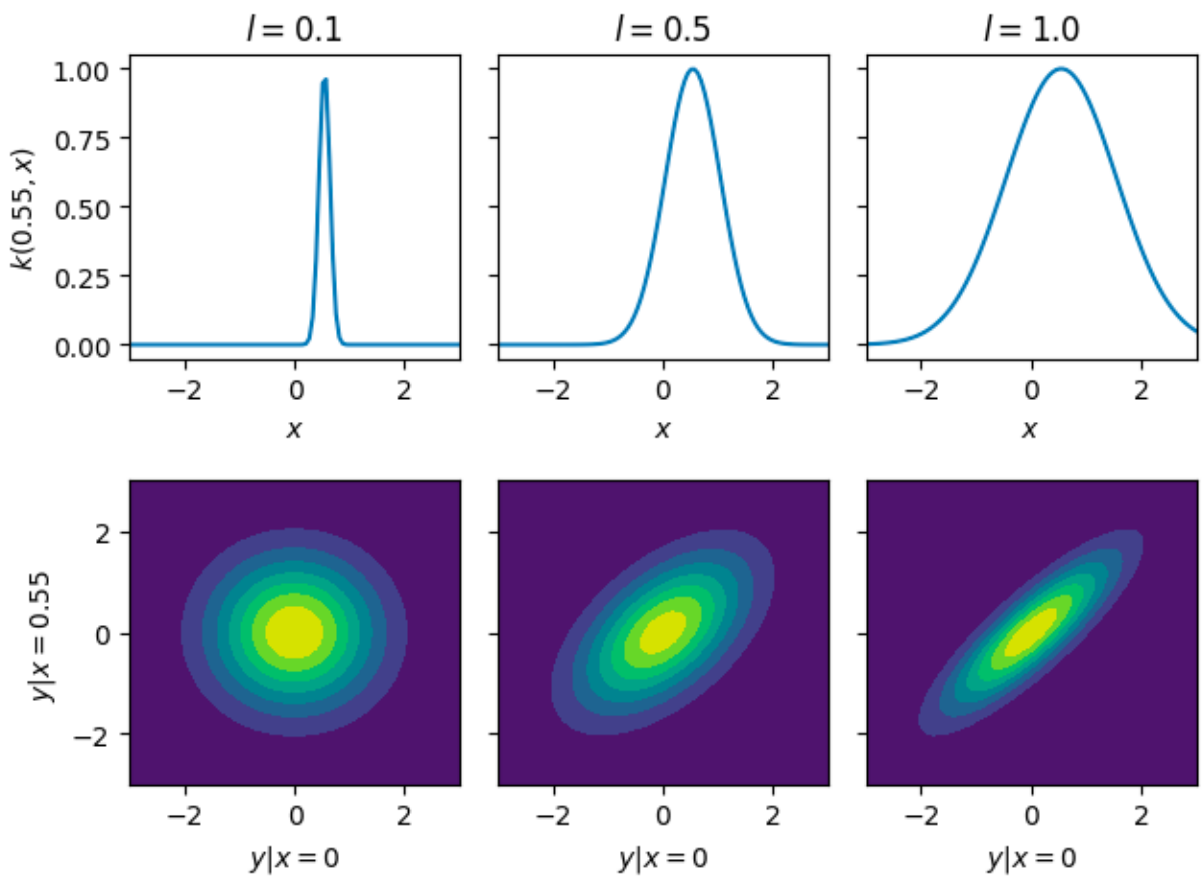
some symmetric and
positive definite
kernel function k

Example: Radial Basis Function Kernel

$$k(\boldsymbol{x}, \boldsymbol{x}') = \exp\left(-\frac{\|\boldsymbol{x}' - \boldsymbol{x}\|^2}{2l^2}\right)$$

length scale

resulting joint distribution of function values of test point and first training point



$p(y_*, y_1 \mid x_*, x_1)$

How to Choose Model?

To choose model (σ^2, kernel function, and its parameters), we can again use marginal likelihood.

Can be computed exactly as for Bayesian ridge regression case (with adapted kernel matrix). We still have:

$$\boldsymbol{t} \mid \boldsymbol{X} \sim \mathrm{N}_n(\boldsymbol{0}, \boldsymbol{K} + \sigma^2 \boldsymbol{I})$$

and therefore log marginal likelihood is given as

$$\begin{aligned}\ln p(\boldsymbol{t} \mid \boldsymbol{X}) &= \ln\left(|2\pi(\boldsymbol{K} + \sigma^2\boldsymbol{I})|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}\boldsymbol{t}^T(\boldsymbol{K} + \sigma^2\boldsymbol{I})^{-1}\boldsymbol{t}\right)\right) \\ &= -\frac{n}{2}\ln 2\pi - \frac{1}{2}\ln|\boldsymbol{K} + \sigma^2\boldsymbol{I}| - \frac{1}{2}\boldsymbol{t}^T(\boldsymbol{K} + \sigma^2\boldsymbol{I})^{-1}\boldsymbol{t}\end{aligned}$$

GPR prediction via conditioning

Prior for any collection of target values $\boldsymbol{t} = (t_1, \dots, t_{N+M})$ given corresponding data matrix is normally distributed

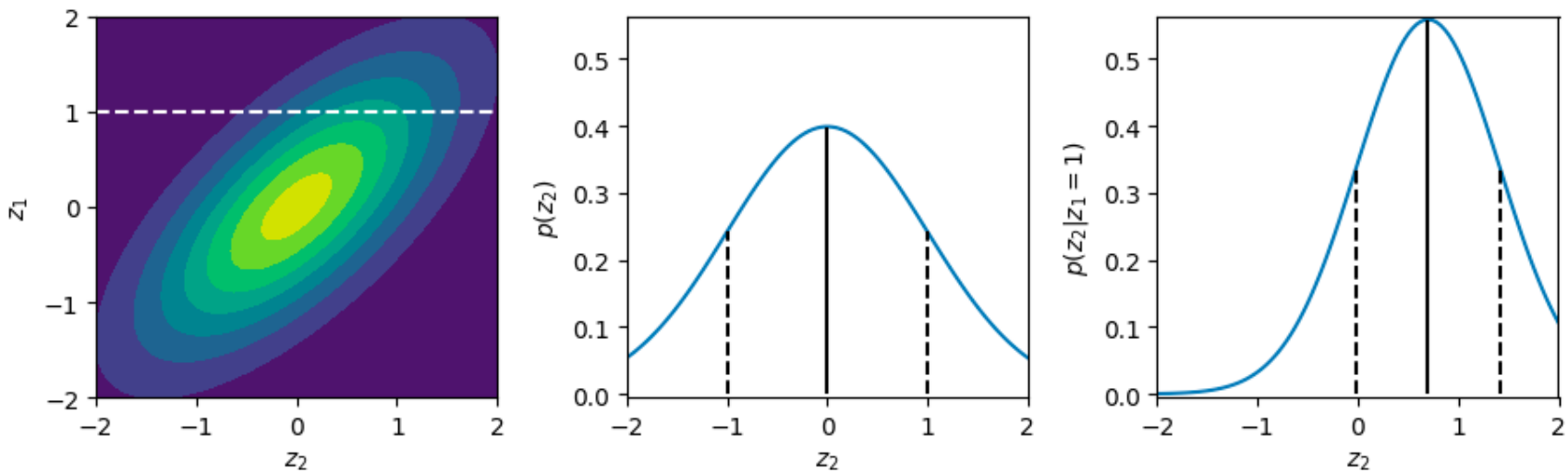
$$\boldsymbol{t} \mid \boldsymbol{X} \sim N_{N+M}(\boldsymbol{0}, \boldsymbol{K} + \sigma^2 \boldsymbol{I})$$

If $\boldsymbol{t_1} = (t_1, \dots, t_N)$ are observed training data, we get predictive distribution for others (test data $\boldsymbol{t_2} = (t_1, \dots, t_M)$) via conditioning

$$\boldsymbol{t}_2 \mid \boldsymbol{t}_1, \boldsymbol{X} \sim ?$$

Conditioning for bivariate normal

Example: $(z_1, z_2) \sim \mathrm{N}_2(\mathbf{0}, [1.0, 0.7; 0.7, 1.0])$



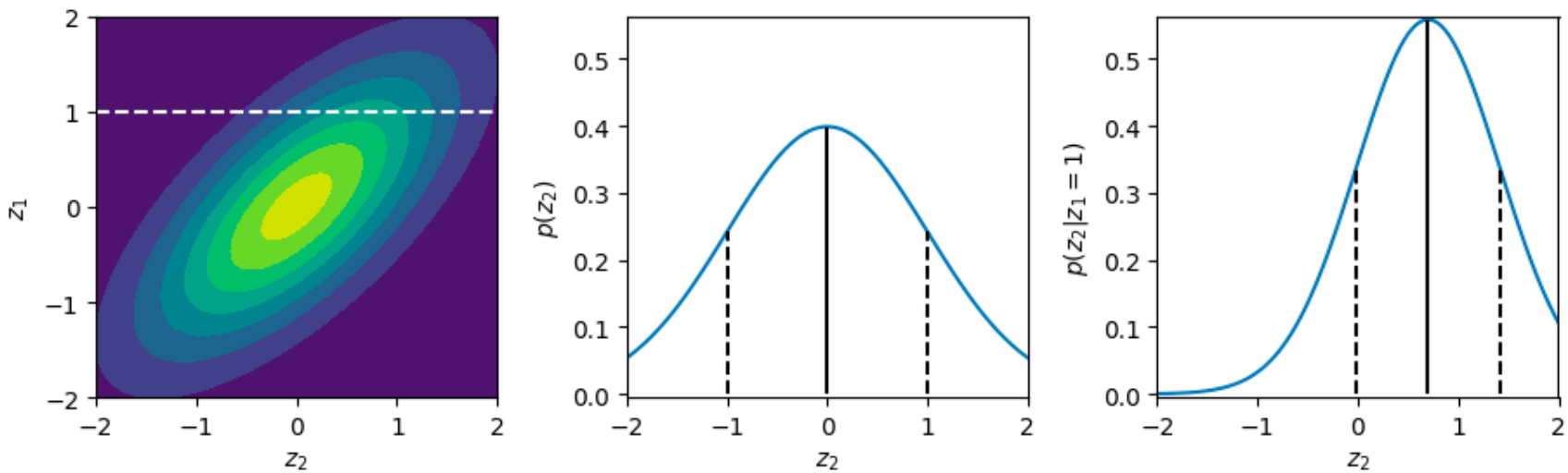
For a normal random vector $(z_1, z_2) \sim \mathrm{N}_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the conditional distribution of z_2 given z_1 is normal with

- mean $\mu_2 + \sigma_{1,2}\sigma_{1,1}^{-1}(z_1 - \mu_1)$
- covariance $\sigma_{2,2} - \sigma_{2,1}\sigma_{1,1}^{-1}\sigma_{1,2}$

with mean vector $\boldsymbol{\mu} = (\mu_1, \mu_2)$ and covariance matrix $\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{1,1} & \sigma_{1,2} \\ \sigma_{2,1} & \sigma_{2,2} \end{bmatrix}$.

Conditioning: general normal vectors

Example: $(z_1, z_2) \sim \mathrm{N}_2(\mathbf{0}, [1.0, 0.7; 0.7, 1.0])$



For a normal random vector $\mathbf{z} \sim \mathrm{N}_{k+l}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ partitioned into $\boldsymbol{z}_1 = (z_1, \dots, z_k)$ and $\boldsymbol{z}_2 = (z_{k+1}, , \dots, z_{k+l})$, the conditional distribution of $\boldsymbol{z}_2$ given $\boldsymbol{z}_1$ is normal with

- mean vector $\boldsymbol{\mu}_2 + \boldsymbol{\Sigma}_{2,1}\boldsymbol{\Sigma}_{1,1}^{-1}(\boldsymbol{z}_1 - \boldsymbol{\mu}_1)$
- covariance matrix $\boldsymbol{\Sigma}_{2,2} - \boldsymbol{\Sigma}_{2,1}\boldsymbol{\Sigma}_{1,1}^{-1}\boldsymbol{\Sigma}_{1,2}$

with the partitioned mean $\boldsymbol{\mu} = (\boldsymbol{\mu}_1, \boldsymbol{\mu}_2)$ and covariance matrix $\boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{1,1} & \boldsymbol{\Sigma}_{1,2} \\ \boldsymbol{\Sigma}_{2,1} & \boldsymbol{\Sigma}_{2,2} \end{bmatrix}$.

Application to GPR prediction

For a normal random vector $\mathbf{z} \sim \mathrm{N}_{k+l}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ partitioned into $\boldsymbol{z}_1 = (z_1, \dots, z_k)$ and $\boldsymbol{z}_2 = (z_{k+1}, , \dots, z_{k+l})$, the conditional distribution of $\boldsymbol{z}_2$ given $\boldsymbol{z}_1$ is normal with

- mean vector $\boldsymbol{\mu}_2 + \boldsymbol{\Sigma}_{2,1}\boldsymbol{\Sigma}_{1,1}^{-1}(\boldsymbol{z}_1 - \boldsymbol{\mu}_1)$
- covariance matrix $\boldsymbol{\Sigma}_{2,2} - \boldsymbol{\Sigma}_{2,1}\boldsymbol{\Sigma}_{1,1}^{-1}\boldsymbol{\Sigma}_{1,2}$

with the partitioned mean $\boldsymbol{\mu} = (\boldsymbol{\mu}_1, \boldsymbol{\mu}_2)$ and covariance matrix $\boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{1,1} & \boldsymbol{\Sigma}_{1,2} \\ \boldsymbol{\Sigma}_{2,1} & \boldsymbol{\Sigma}_{2,2} \end{bmatrix}$.

In our case $\boldsymbol{t} \mid \boldsymbol{X} \sim \mathrm{N}_{N+M}(\mathbf{0}, \boldsymbol{K} + \sigma^2 \boldsymbol{I})$ and want to know $\boldsymbol{t}_2 \mid \boldsymbol{t}_1, \boldsymbol{X}$.

Setting $\boldsymbol{z}_1 = \boldsymbol{t}_1, \boldsymbol{z}_2 = \boldsymbol{t}_2$ we have

- $\boldsymbol{\mu}_1 = \boldsymbol{\mu}_2 = \mathbf{0}$
- $\boldsymbol{\Sigma}_{1,1} = \boldsymbol{K}_{1,1} + \sigma^2 \boldsymbol{I}$
- $\boldsymbol{\Sigma}_{2,2} = \boldsymbol{K}_{2,2} + \sigma^2 \boldsymbol{I}$
- $\boldsymbol{\Sigma}_{1,2} = \boldsymbol{\Sigma}_{2,1}^T = \boldsymbol{K}_{1,2} = \boldsymbol{K}_{2,1}^T$

Resulting for $\boldsymbol{t}_2 \mid \boldsymbol{t}_1$ in conditional

- mean $\boldsymbol{K}_{2,1}\left(\boldsymbol{K}_{1,1} + \sigma^2 \boldsymbol{I}\right)^{-1} \boldsymbol{t}_1$
- covar. $\boldsymbol{K}_{2,2} + \sigma^2 \boldsymbol{I} - \boldsymbol{K}_{2,1}\left(\boldsymbol{K}_{1,1} + \sigma^2 \boldsymbol{I}\right)^{-1} \boldsymbol{K}_{1,2}$

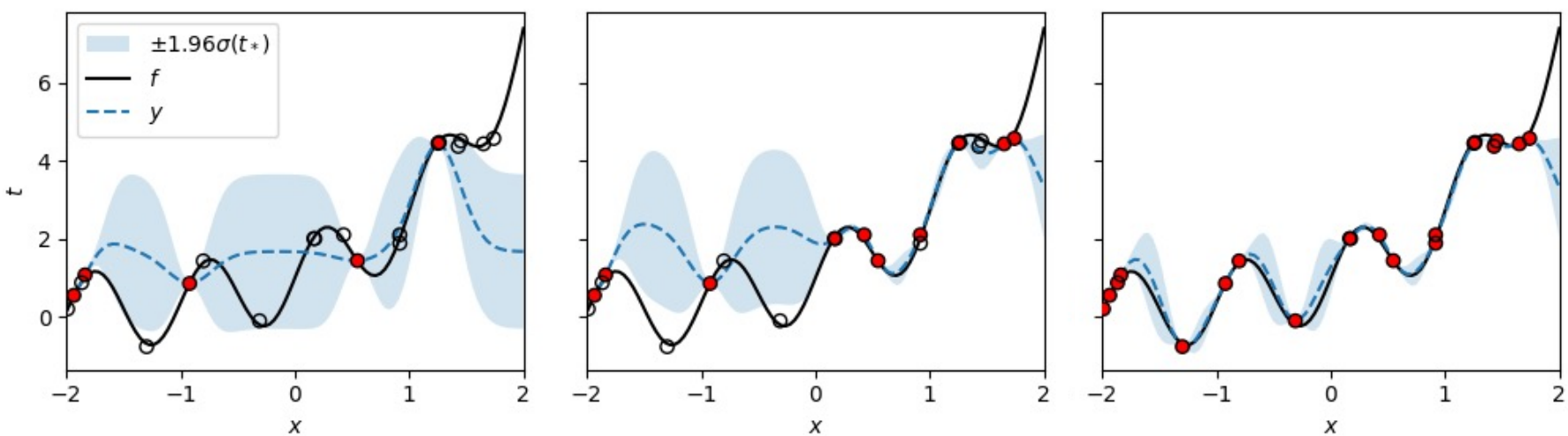
GPR prediction via conditioning

Any collection of target values $\boldsymbol{t} = (t_1, \dots, t_{N+M})$ given input matrix $\boldsymbol{X}$ has multivariate normal prior

$$\boldsymbol{t} \mid \boldsymbol{X} \sim N_{N+M}(\boldsymbol{0}, \boldsymbol{K} + \sigma^2 \boldsymbol{I})$$

If $\boldsymbol{t_1} = (t_1, \dots, t_N)$ are observed training data, we get predictive distribution for others (test data $\boldsymbol{t_2} = (t_1, \dots, t_M)$) via conditioning

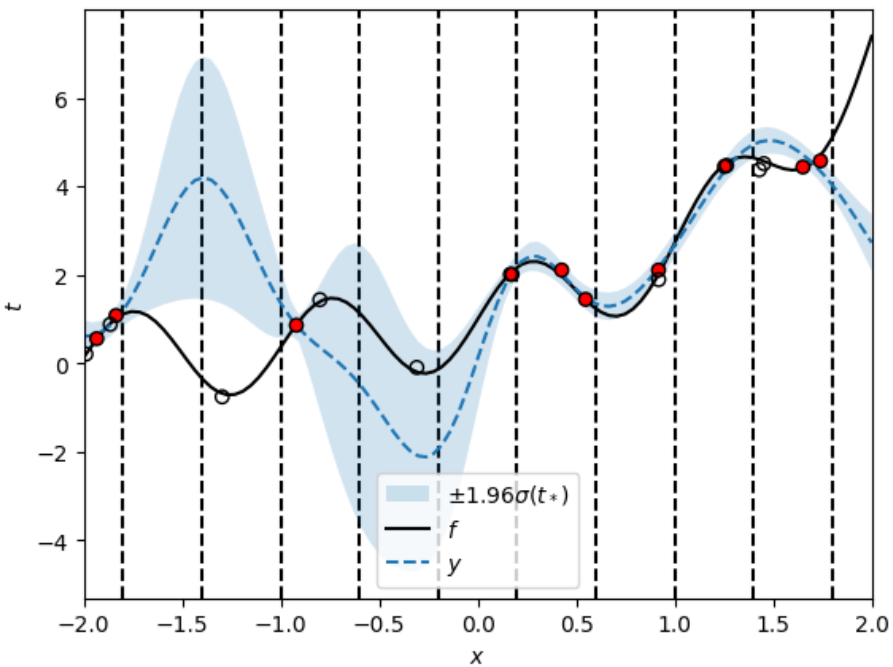
$$\boldsymbol{t_2} \mid \boldsymbol{t_1}, \boldsymbol{X} \sim \mathrm{N}_M \left(\boldsymbol{K}_{2,1} \left(\boldsymbol{K}_{1,1} + \sigma^2 \boldsymbol{I} \right)^{-1} \boldsymbol{t_1}, \boldsymbol{K}_{2,2} + \sigma^2 \boldsymbol{I} - \boldsymbol{K}_{2,1} (\boldsymbol{K}_{1,1} + \sigma^2 \boldsymbol{I})\, \boldsymbol{K}_{1,2} \right)$$



Comparison to parametric model

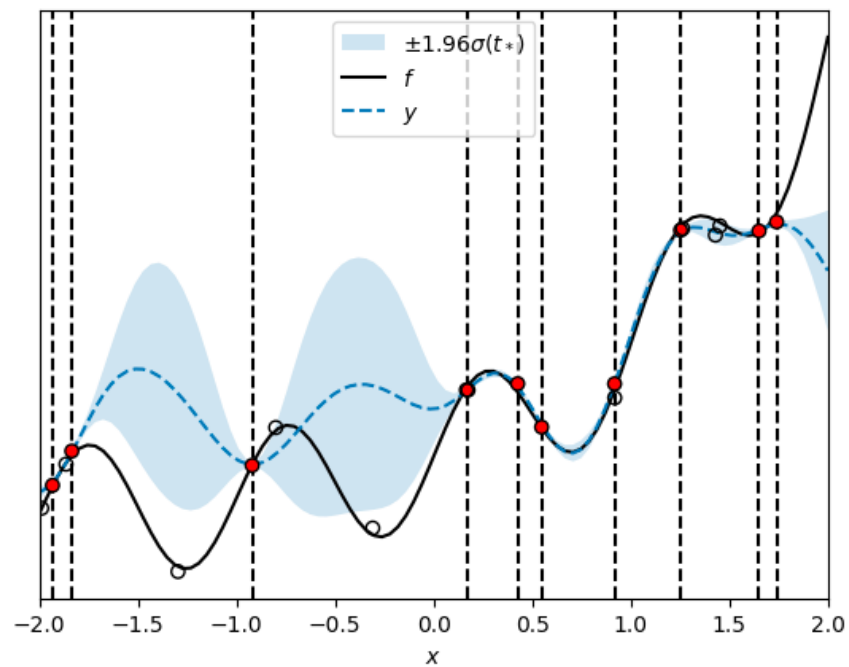
Can rewrite GPR predictions as linear model with features $\phi_i(x) = k(x_i, x)$

Bayesian Ridge Regression



$$y(\boldsymbol{x}_*) = \boldsymbol{\phi}(x_*)^T(\boldsymbol{\Phi}^T\boldsymbol{\Phi} + \sigma^2/\alpha^2\boldsymbol{I})^{-1}\boldsymbol{t}_1$$

Gaussian Process Regression



$$y(\boldsymbol{x}_*) = \boldsymbol{\phi}(x_*)^T(\boldsymbol{\Phi} + \sigma^2\boldsymbol{I})^{-1}\boldsymbol{t}_1$$

Key points: Gaussian Process Regression

- GPR generalises Bayesian ridge regression by directly defining prior on function values $\boldsymbol{y} \mid \boldsymbol{X} \sim N(\boldsymbol{0}, \boldsymbol{K})$
- …where kernel function k determines kernel matrix entries $\boldsymbol{K}_{i,j} = k(\boldsymbol{x}_i, \boldsymbol{x}_j)$
- Popular choice: radial basis function kernel $k(\boldsymbol{x}, \boldsymbol{x}') = \exp(-\|\boldsymbol{x} - \boldsymbol{x}'\|^2/2l^2)$
- GPR allows again model selection via marginal likelihood with key terms related to complexity $-\frac{1}{2}\log|\boldsymbol{K} + \sigma^2\boldsymbol{I}|$ and fit $-\frac{1}{2}\boldsymbol{t}^T(\boldsymbol{K} + \sigma^2\boldsymbol{I})^{-1}\boldsymbol{t}$
- GPR prediction based on conditioning prior for union of training and test inputs on the observed target values
$$\boldsymbol{t_2} \mid \boldsymbol{t_1}, \boldsymbol{X} \sim \mathrm{N}_M\left(\boldsymbol{K}_{2,1}\left(\boldsymbol{K}_{1,1} + \sigma^2\boldsymbol{I}\right)^{-\boldsymbol{1}}\boldsymbol{t_1}, \boldsymbol{K}_{2,2} + \sigma^2\boldsymbol{I} - \boldsymbol{K}_{2,\boldsymbol{1}}(\boldsymbol{K}_{1,1} + \sigma^2\boldsymbol{I})\,\boldsymbol{K}_{1,2}\right)$$
- Prediction for single test point $\boldsymbol{x}$ corresponds to linear combination $y(\boldsymbol{x}) = \boldsymbol{\phi}(\boldsymbol{x})^T\boldsymbol{w}$ with weights $\boldsymbol{w} = \left(\boldsymbol{K}_{1,1} + \sigma^2\boldsymbol{I}\right)^{-\boldsymbol{1}}\boldsymbol{t_1}$ and basis transformation
$$\boldsymbol{\phi}(\boldsymbol{x}) = \left(k(\boldsymbol{x}_1, \boldsymbol{x}), \dots, k(\boldsymbol{x}_N, \boldsymbol{x})\right)$$

Laboratory (Week 11)

Implement and discuss

- ❑ Bayesian ridge regression
- ❑ Radial basis function kernel
- ❑ Gaussian process regression
- ❑ Model selection via marginal likelihood

What will we learn in Week 12

Part B in Module 6

- ☐ …finishing our advanced topics
- ☐ Unit revision